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Iterative Symmetric Centipede Game: An Evolutionary Approach

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1 Introduction

In this work we study games derived from the *Centipede* game, an extensive-form game of perfect information, as well as normal-form variants of the game, in particular the *Iterated Symmetric Centipede* (ISC), which arises through the following procedure.

First, using a standard construction, we obtain a normal-form game that is equivalent to the classic tree-form Centipede. To remove the asymmetry between the first and the second player in the original Centipede, we modify the game by randomizing the choice of which player moves first. This yields the *One-Round Symmetric Centipede* (OSC). However, the main focus of this thesis is on the *Iterated Symmetric Centipede* (ISC), which consists of T successive rounds of the OSC. We assume that in each round of the ISC the two players have memory of the moves and outcomes of the previous rounds. This introduces the possibility of using strategies that depend on the history of the game. This thesis studies the effect of such strategies on promoting cooperation and punishing defection.

Within the interpretation of the iterated Centipede game adopted at the level of rounds, where each round constitutes a complete single-player Centipede "tournament" and strategies are updated only between rounds. From this perspective, it is natural and logical to transfer strategies from the Iterated Prisoner's Dilemma (IPD), such as Tit-for-Tat and Grim Trigger, to the ISC. To determine behavior in the current round, such strategies use the opponent's actions in previous rounds, rather than reacting exclusively to the decisions made within the same round.

The structure of the analysis is as follows. Section 2 presents a brief review of the classic Centipede, the one-round normal-form symmetric Centipede, and the ISC. In Section 3, an ISC game with three (3) strategies is studied theoretically and implemented. Specifically, the theoretical analysis includes Nash Equilibria (NE), replicator dynamics, Evolutionarily Stable Strategies (ESS), and Markov analysis. In Section 4, the theoretical analysis and the implementation are repeated for an ISC game with five (5) strategies. Section 5 studies a new evolutionary dynamic in which the ISC game is played with three (3) strategies between two populations, as well as the evolutionarily stable strategies of this game.

The work also includes an appendix, which describes the functions and code used to implement the games under study in MATLAB.

2 Fundamental Concepts

2.1 The Classic Centipede Game

The classic centipede game, as well as all of its variants, has the following basic characteristics:

1. It is an extensive-form game with perfect information.
2. The game tree has the characteristic "centipede" shape, that is, it consists of a path of M vertices, where each vertex of the path is also connected to a leaf.
3. The terminal vertices are $M+1, M+2, \ldots, 2M$. Vertex $M+m$ has payoffs (a_m, b_m).
4. The payoffs satisfy:
$$a_1 > a_2, \quad a_3 > a_4, \quad a_5 > a_6, \ldots$$
and
$$b_1, b_2 > b_3, \quad b_4 > b_5, \quad b_6 > b_7, \ldots$$

The game is played as follows:

1. Player P_1 plays at odd time steps and P_2 at even ones.
2. The total payoff at round t is
$$c_t = a_t + b_t = f(t),$$
where f is an increasing function of t.
3. If the player whose turn it is terminates the game, they receive a share pc_t, while the other player receives $(1-p)c_t$, where
$$p \in \left(\frac{1}{2}, 1\right].$$

In this work it is always assumed that
$$f(t) = t.$$

At every non-terminal vertex, the player whose turn it is can either terminate the game or continue it to the next stage. The total benefit increases as the duration of the game grows, while the player who terminates the game receives a larger share of the total payoff.

We now define a family of strategies
$$\rho_1, \rho_2, \ldots, \rho_M$$
as follows:

ρ_m : "I terminate the game as soon as it reaches or passes stage m and it is my turn to play."

Let the two players be P_1 and P_2. For $i \in \{1,2\}$ and $m, n \in \{1, \ldots, M\}$ we define:

$$Q_i(\rho_m, \rho_n) = \text{the payoff of } P_i \text{ when } P_1 \text{ plays } \rho_m \text{ and } P_2 \text{ plays } \rho_n.$$

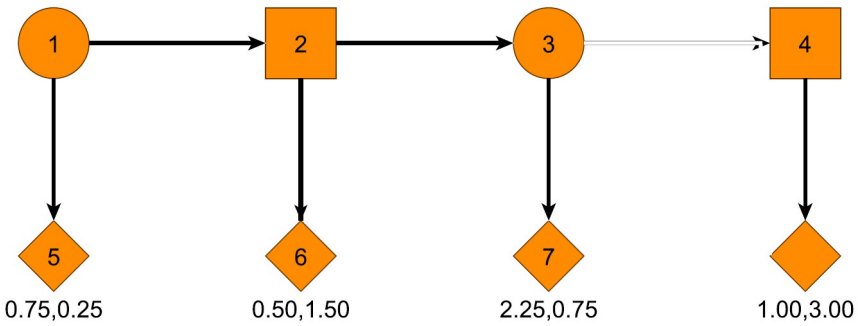


Figure 1: Centipede Game (figure from the thesis of Lazaridis-Kechagias)

2.2 One-Round Symmetric Centipede (OSC)

Using a standard procedure, the tree-form Centipede is converted into a normal-form bimatrix game, which we call the *One-Round Symmetric Centipede* (OSC). The assumptions of the classic game are retained: the total payoff at time t is t, P_1 receives pt and P_2 receives $(1-p)t$.

We define the matrices $A^{(1)}$ and $A^{(2)}$ as:

$$A^{(i)}_{mn} = \text{the payoff of } P_1 \text{ when } P_1 \text{ plays } \rho_m,\ P_2 \text{ plays } \rho_n \text{ and } P_i \text{ plays first.}$$

If P_1 plays first, then:

$$A^{(1)}_{ij} = \begin{cases} p\,\min(i,j), & i < j, \\ (1-p)\,\min(i,j), & i > j, \end{cases}$$

when i is odd and j is even. In addition:

$$A^{(1)}_{ij} = A^{(1)}_{i,j+1}$$

when both are odd,

$$A^{(1)}_{ij} = A^{(1)}_{i+1,j}$$

when both are even, and

$$A^{(1)}_{ij} = A^{(1)}_{i+1,j+1}$$

when i is even and j is odd. We define the expected payoff matrix:

$$A = \frac{1}{2}\left(A^{(1)} + A^{(2)}\right).$$

Ignoring the factor $\frac{1}{2}$, which does not affect the subsequent results, we finally obtain the matrix:

$$A = \begin{bmatrix} 1 & 3p & 3p & 3p & \cdots & 3p \\ 3(1-p) & 2 & 5p & 5p & \cdots & 5p \\ 3(1-p) & 5(1-p) & 3 & 7p & \cdots & 7p \\ 3(1-p) & 5(1-p) & 7(1-p) & 4 & \cdots & 9p \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 3(1-p) & 5(1-p) & 7(1-p) & 9(1-p) & \cdots & M \end{bmatrix}.$$

2.3 Iterated Symmetric Centipede (ISC)

We now introduce the *Iterated Symmetric Centipede* (ISC) game. The ISC is obtained from the OSC in the usual way used for repeated games: we play the OSC for T successive rounds.

The feature that qualitatively distinguishes the ISC from the OSC is that players can use strategies with memory. That is, the OSC strategy used in round t may depend on the moves of previous rounds. In this sense, the ISC is related to the OSC as the Iterated Prisoner's Dilemma (IPD) is related to the Prisoner's Dilemma (PD).

More specifically, we define the ISC as a game between two players, each of whom chooses one strategy. Within the scope of this thesis, the available strategies are finite in number and well defined. After the strategies are chosen, the players play a sequence of T OSC rounds.

We define the payoff matrix C as:

$$C_{ij} = \text{the payoff of a player using strategy } i \text{ when playing against a player using strategy } j.$$

The matrix C is computed from the matrix A of the OSC, and the method of computation will be presented in the next section.

It should also be noted that, besides pure strategies, players can play mixed strategies σ, where σ is a probability vector whose i-th element expresses the probability with which the i-th strategy is played.

3 Implementation of the Game with 3 Strategies

3.1 Strategies and Payoffs

In this section of the thesis, the following three strategies are chosen for the theoretical analysis and implementation of the game:

1. $\sigma_{\frac{M}{2}}$ or All-$\frac{M}{2}$: in every OSC the strategy $\rho_{\frac{M}{2}}$ is played.
2. σ_1 or All-1: in every OSC the strategy ρ_1 is played.
3. σ_G or Grim: in the first round ρ_M is played, and it continues to be played as long as the opponent has played ρ_M in all previous rounds. If in some round the opponent plays ρ_m with $m < M$, then from that point on ρ_1 is played permanently.

Consequently, the ISC is defined as a game between two players, each of whom chooses one of the strategies

$$\text{All - } \frac{M}{2}, \quad \text{All - 1}, \quad \text{Grim}$$

For these particular strategies, some examples of computing elements of the matrix C follow:

- C_{11} : is the payoff of the player who has chosen the strategy $\sigma_{\frac{M}{2}}$ when playing against a player who has chosen the same strategy. In this case, since both play $\rho_{\frac{M}{2}}$ for T rounds, the payoff will be $TA_{\frac{M}{2}\frac{M}{2}}$.
- C_{13} : is the payoff of the player who has chosen the strategy $\sigma_{\frac{M}{2}}$ when playing against a player who has chosen the strategy σ_G. In this case, since this player plays strategy $\rho_{\frac{M}{2}}$ and the other plays ρ_M in the first round of the game, he obtains payoff $A_{\frac{M}{2}M}$. In the remaining rounds, however, that is in $(T-1)$ rounds, the second player now plays strategy ρ_1, so the term $(T-1)A_{\frac{M}{2}1}$ is added to the payoff of the player in question.
- C_{32} : is the payoff of the player who has chosen the strategy σ_G when playing against a player who has chosen the strategy σ_1. In this case, since this player plays strategy ρ_M in the first round and the other plays strategy ρ_1, the player obtains payoff A_{M1} from the first round. In the remaining rounds he switches to strategy ρ_1, so the term $(T-1)A_{11}$ is added to the payoff.

The remaining elements of the matrix C are obtained in a similar way. Finally, the payoff matrix for player P_1 is

$$C = \begin{bmatrix} TA_{\frac{M}{2}\frac{M}{2}} & TA_{\frac{M}{2}1} & A_{\frac{M}{2}M} + (T-1)A_{\frac{M}{2}1} \\ TA_{1\frac{M}{2}} & TA_{11} & A_{1M} + (T-1)A_{11} \\ A_{M\frac{M}{2}} + (T-1)A_{1\frac{M}{2}} & A_{M1} + (T-1)A_{11} & TA_{MM} \end{bmatrix},$$

that is

$$C = \begin{bmatrix} \frac{TM}{2} & 3T(1-p) & (M+1)p + (T-1)3(1-p) \\ 3Tp & T & 3p + (T-1) \\ (M+1)(1-p) + (T-1)3p & 3(1-p) + (T-1) & TM \end{bmatrix}.$$

The bimatrix game is $D = (C, C^T)$, where C^T is the payoff of player P_2.

3.2 Nash Equilibria

Definition 3.2.1 A pair of strategies $(\hat{\sigma_1}, \hat{\sigma_2})$ is called a Nash Equilibrium (NE) in a two-player game with payoffs Q_1, Q_2 if and only if

$$\forall \sigma_1 : Q_1(\hat{\sigma}_1, \hat{\sigma}_2) \geq Q_1(\sigma_1, \hat{\sigma}_2) \quad \text{and} \quad \forall \sigma_2 : Q_2(\hat{\sigma}_1, \hat{\sigma}_2) \geq Q_2(\hat{\sigma}_1, \sigma_2) \tag{1}$$

We now give a family of NE for the ISC. The strategies

Proposition 3.2.2 For sufficiently large T and for $p \geq \frac{2}{3}$, the following is a symmetric NE.

$$((0,1,0),(0,1,0))$$

Proof The payoff of player P_1 for the strategy pair $((0,1,0),(0,1,0))$ is:

$$Q_1((0,1,0),(0,1,0)) =$$

$$\begin{bmatrix}0 & 1 & 0\end{bmatrix}\begin{bmatrix}\frac{TM}{2} & 3T(1-p) & (M+1)p+(T-1)3(1-p)\\ 3Tp & T & 3p+(T-1)\\ (M+1)(1-p)+(T-1)3p & 3(1-p)+(T-1) & TM\end{bmatrix}\begin{bmatrix}0\\1\\0\end{bmatrix} =$$

$$= \begin{bmatrix}0 & 1 & 0\end{bmatrix}\begin{bmatrix}3T(1-p)\\ T\\ 3(1-p)+(T-1)\end{bmatrix} = T$$

Suppose that P_1 switches to the strategy $\tau = (\alpha, 1-\alpha-b, b)$

$$Q_1(\tau,(0,1,0)) =$$

$$\begin{bmatrix}\alpha & 1-\alpha-b & b\end{bmatrix}\begin{bmatrix}\frac{TM}{2} & 3T(1-p) & (M+1)p+(T-1)3(1-p)\\ 3Tp & T & 3p+(T-1)\\ (M+1)(1-p)+(T-1)3p & 3(1-p)+(T-1) & TM\end{bmatrix}\begin{bmatrix}0\\1\\0\end{bmatrix} =$$

$$= \begin{bmatrix}\alpha & 1-\alpha-b & b\end{bmatrix}\begin{bmatrix}3T(1-p)\\ T\\ 3(1-p)+(T-1)\end{bmatrix} =$$

$$= 3T(1-p)\alpha + T(1-\alpha-b) + 3b(1-p) + b(T-1)$$

The difference in payoffs is

$$\delta Q_1 = Q_1((0,1,0),(0,1,0)) - Q_1(\tau,(0,1,0)) =$$

$$= -3T\alpha + 3T\alpha p + T\alpha + Tb - 3b + 3bp - bT + b$$

$$= T\alpha(-2+3p) - 3b + 3bp + b$$

For sufficiently large T, the sign of δQ depends on the term $\alpha T(-2+3p)$, which is positive for $p > \frac{2}{3}$, hence $\delta Q_1 > 0$. Following the same procedure, we also obtain $\delta Q_2 > 0$. Therefore, $((0,1,0),(0,1,0))$ is a NE for $p \geq \frac{2}{3}$ ■

Proposition 3.2.3 For $M \in \{3,4,\dots\}$, $p \geq \frac{2}{3}$ and for sufficiently large T, the following is a symmetric NE

$$((1,0,0),(1,0,0))$$

Proof Similarly to before,

$$Q_1((0,1,0),(0,1,0)) =$$

$$\begin{bmatrix}1 & 0 & 0\end{bmatrix}\begin{bmatrix}\frac{TM}{2} & 3T(1-p) & (M+1)p+(T-1)3(1-p)\\ 3Tp & T & 3p+(T-1)\\ (M+1)(1-p)+(T-1)3p & 3(1-p)+(T-1) & TM\end{bmatrix}\begin{bmatrix}1\\0\\0\end{bmatrix} = \frac{TM}{2}$$

Suppose that P_1 switches to the strategy $\tau = (\alpha, 1-\alpha-b, b)$. Then

$$
\begin{aligned}
&Q_1(\tau,(1,0,0)) = \\
&= \begin{bmatrix} a & b & 1-a-b \end{bmatrix} \begin{bmatrix} \frac{TM}{2} & 3T(1-p) & (M+1)p+(T-1)3(1-p) \\ 3Tp & T & 3p+(T-1) \\ (M+1)(1-p)+(T-1)3p & 3(1-p)+(T-1) & TM \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \\
&= ((M+1)(p-1)-p(3T-3))(a+b-1)+\frac{aMT}{2}+3Tbp
\end{aligned}
$$

Consequently:

$$\delta Q = Q_1((1,0,0),(1,0,0)) - Q_1(\tau,(1,0,0)) = T(1-a)(\tfrac{M}{2}-3p)+(M-Mp-1+4p)(1-a-b)$$

For sufficiently large T, the sign of the quantity δQ is determined by the first-order terms in T, that is:

$$T(1-a)(\frac{M}{2}-3p) \geq 0$$

which holds for every $M \geq 6p$. Similarly, under the same conditions we obtain $\delta Q_2 > 0$. Therefore $((1,0,0),(1,0,0))$ is a NE for $p \geq \frac{2}{3}$. ■

Proposition 3.2.4 For $M \in \{1,2,\dots\}$ and for sufficiently large T, the following is a symmetric NE

$$((0,0,1),(0,0,1))$$

Proof Similarly to before,

$$
\begin{aligned}
&Q_1((0,0,1),(0,0,1)) = \\
&\begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{TM}{2} & 3T(1-p) & (M+1)p+(T-1)3(1-p) \\ 3Tp & T & 3p+(T-1) \\ (M+1)(1-p)+(T-1)3p & 3(1-p)+(T-1) & TM \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = TM
\end{aligned}
$$

Suppose that P_1 switches to the strategy $\tau = (\alpha, 1-\alpha-b, b)$. Then,

$$
\begin{aligned}
&Q_1(\tau,(0,1,0)) = \\
&\begin{bmatrix} a & b & 1-a-b \end{bmatrix} \begin{bmatrix} \frac{TM}{2} & 3T(1-p) & (M+1)p+(T-1)3(1-p) \\ 3Tp & T & 3p+(T-1) \\ (M+1)(1-p)+(T-1)3p & 3(1-p)+(T-1) & TM \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \\
&= b(T+3p-1)-a[(3T-3)(p-1)-p(M+1)]-MT(a+b-1)
\end{aligned}
$$

Consequently,

$$\delta Q = Q_1((0,0,1),(0,0,1)) - Q_1(\tau,(0,0,1)) = T(-b+3ap+Ma+Mb)+(-Mpa-4ap-3a-3bp+b)$$

For sufficiently large T, the sign of the quantity δQ is determined by the first-order terms in T, that is:

$$
\begin{aligned}
T(-b+3ap+Ma+Mb) &> T(-b+Ma+Mb) \\
&= T[Ma+(M-1)b]
\end{aligned}
$$

Since $a, b \in (0,1)$ and $M \geq 1$ (otherwise the centipede game would be trivial), the first-order term in T will always be positive. As before, in the same way we again obtain $\delta Q_2 > 0$. Therefore, we conclude that $(0,0,1)$ is indeed a NE ■

Proposition 3.2.5 For $M \in \{6,7,\dots\}$, for $p > \frac{2}{3}$ and for sufficiently large T, the following is a symmetric NE

$$\left(\left(\frac{6p-4}{M-4}, \frac{M-6p}{M-4}, 0\right), \left(\frac{6p-4}{M-4}, \frac{M-6p}{M-4}, 0\right)\right)$$

Proof We examine the form $((a, 1-a, 0), (c, 1-c, 0))$. In a similar way as above we obtain:

$$Q_1(a,c) = \frac{TM}{2}ac + (1-c)3T(1-p)a + 3Tcp(1-a) + (1-a)T(1-c) = 0$$
$$Q_2(a,c) = \frac{TM}{2}ac + (1-c)3Tpa + 3Tc(1-p)(1-a) + (1-a)T(1-c) = 0$$

and solving

$$\frac{dQ_1}{da} = \frac{TMc}{2} + (1-c)3T(1-p) - 3Tcp - T(1-c)$$
$$\frac{dQ_2}{dc} = \frac{TMa}{2} + (1-a)3T(1-p) - 3Tap - T(1-a)$$

we get

$$a = c = \frac{6p-4}{M-4}$$

Admissible points are those for which the constraints $M \in \{6, 7, \dots\}$ and $p > \frac{2}{3}$ hold. For $p = \frac{2}{3}$ the point is $(0, 1, 0)$, for which we have already proved that it is a symmetric NE ■

Proposition 3.2.6 For $M \in \{3, 4, \dots\}$,$p > \frac{2}{3}$ and for sufficiently large T, the following is a symmetric NE

$$\left(\left(0, \frac{T+3p-MT-1}{T-MT+1}, -\frac{3p-2}{T-MT+1}\right), \left(0, \frac{T+3p-MT-1}{T-MT+1}, -\frac{3p-2}{T-MT+1}\right)\right)$$

Proof We examine the form $((0, a, 1-a), (0, c, 1-c))$. In a similar way as above, we obtain:

$$Q_1(a,c) = Tac + (3p + (T-1))(1-c)a + TM(1-c)(1-a) + (3(1-p) + (T-1))c(1-a)$$

$$Q_2(a,c) = Tac + (3(1-p) + (T-1))(1-c)a + (3p + (T-1))c(1-a) + (1-a)MT(1-c)$$

and solving

$$\frac{dQ_1}{da} = Tc + (3p + (T-1))(1-c) - TM(1-c) - c(3(1-p) + (T-1))$$

$$\frac{dQ_2}{dc} = Ta + (3p + (T-1))(1-a) - TM(1-a) - a(3(1-p) + (T-1))$$

we get

$$a = c = \frac{T+3p-MT-1}{T-MT+1}$$

Admissible points are those for which the constraints $M \in \{3, 4, \dots\}$ and $p > \frac{2}{3}$ hold. For $p = \frac{2}{3}$ the point is $(0, 1, 0)$, for which we have already proved that it is a symmetric NE ■

Proposition 3.2.7 For and for sufficiently large T, the following is a symmetric NE

$$(\tau, \tau)$$

where

$$\tau = \left(\frac{2(3T + 4p - MT + Mp - 3Tp - 3)}{2M + 6T - 3MT - 4}, 0, \frac{2M - 8p - MT - 2Mp + 6Tp + 2}{2M + 6T - 3MT - 4} \right)$$

Proof We examine the form $((a, 0, 1-a), (c, 0, 1-c))$. In a similar way as above, we obtain:

$$Q_1(a,c) = \frac{TM}{2}ac + (1-c)a\left((M+1)p + 3(T-1)(1-p)\right) + (1-a)(1-c)TM + c(1-a)\left((M+1)(1-p) + 3(T-1)p\right)$$

$$Q_2(a,c) = \frac{TM}{2}ac + (1-a)c\left((M+1)p + 3(T-1)(1-p)\right) + (1-a)(1-c)TM + (1-c)a\left((M+1)(1-p) + 3(T-1)p\right)$$

and solving

$$\frac{dQ_1}{da} = \frac{cTM}{2} + (1-c)\left((M+1)p + 3(T-1)(1-p)\right) - (1-c)TM - c\left((M+1)(1-p) + 3(T-1)p\right)$$

$$\frac{dQ_2}{dc} = \frac{aTM}{2} + (1-a)\left((M+1)p + 3(T-1)(1-p)\right) - (1-a)TM - a\left((M+1)(1-p) + 3(T-1)p\right)$$

we get

$$a = c = \frac{2(3T + 4p - MT + Mp - 3Tp - 3)}{2M + 6T - 3MT - 4}$$

Admissible points are those for which the constraints $M \in \{3, 4, \dots\}$ and $p > \frac{2}{3}$ hold ■

3.3 Replicator Dynamics

We consider an evolutionary game involving a large population of players, who play the ISC as the base game and use the strategies $\sigma_{\frac{M}{2}}$, σ_1 and σ_G.

Let x_1 be the fraction of players using the strategy $\sigma_{\text{All-M}}$, x_2 the fraction of players using the strategy $\sigma_{\text{All-1}}$, and x_3 the fraction of players using the strategy σ_G. The variables x_m are assumed to evolve according to the replicator dynamics (*replicator dynamics*).

$$\frac{dx_i}{dt} = x_m((Cx)_i - x^T Cx) \tag{2}$$

where

$$\frac{dx_1}{dt} = x_1[\frac{TM}{2}x_1 + 3T(1-p)x_2 + ((M+1)p + 3(T-1)(1-P))x_3 - (\frac{TM}{2}x_1^2 + 3Tx_1x_2 + (M+3T-2)x_1x_3 + Tx_2^2 + (2T+1)x_2x_3 + TMx_3^2)]$$

$$\frac{dx_2}{dt} = x_2[3Tpx_1 + Tx_2 + (3p + (T-1))x_3 - (\frac{TM}{2}x_1^2 + 3Tx_1x_2 + (M+3T-2)x_1x_3 + Tx_2^2 + (2T+1)x_2x_3 + TMx_3^2)]$$

$$\frac{dx_3}{dt} = x_3[((M+1)(1-p)+3(T-1)p)x_1 + (3(1-p)+(T-1))x_2 + TMx_3 - (\frac{TM}{2}x_1^2 + 3Tx_1x_2 + (M+3T-2)x_1x_3 + Tx_2^2 + (2T+1)x_2x_3 + TMx_3^2)]$$

To find equilibrium points, the following system must be solved:

$$\forall m : 0 = x_m \left((Cx)_m - x^T Cx\right) \tag{3}$$

The following points are obviously solutions of equation (3), which is why they are examined.

Proposition 3.3.1 For every $M \in \{3, 4, \dots\}$ and every T, the following are admissible equilibrium points.

$$(0, 1, 0)$$
$$(1, 0, 0)$$
$$(0, 0, 1)$$

Proof Since they are roots of equation (3) and the necessary constraints for them to represent fractions hold, the points are admissible equilibrium points ■

Proposition 3.3.2 For every $M \in \{6, 7...\}$, $p \in (\frac{2}{3}, 1)$ and every T, the following is an admissible equilibrium point.

$$\left(\frac{6p-4}{M-4}, \frac{M-6p}{M-4}, 0\right)$$

Proof For it to be admissible, the following must hold

$$x_1 + x_2 + x_3 = 1$$
$$x_1, x_2, x_3 > 0$$

For the equality constraint we have:

$$\frac{6p-4}{M-4} + \frac{M-6p}{M-4} + 0 = 1$$
$$6p - 4 + M - 6p = M - 4$$
$$6p - 4 + M = 6p - 4 + M \quad \text{(obviously true)}$$

Since $M \in \{6, 7, ..\}$, we have $M > 4$, so, as the denominator is positive, we must have:

$$6p - 4 > 0 \ \ \& \ \ M - 6p > 0$$
$$p > \frac{2}{3} \ \ \& \ \ M > 6p$$

Since for $p = 1$ we need $M > 6$, and $p \in (\frac{2}{3}, 1)$, it also holds that for $M \in \{6, 7...\}$ the fractions are positive quantities. Hence, it will be an equilibrium point ■

Proposition 3.3.3 For every $M \in \{3, 4, \dots\}$, $p \in (\frac{2}{3}, 1)$ and every T, the following is an admissible equilibrium point.

$$\left(0, \frac{T + 3p - MT - 1}{T - MT + 1}, -\frac{3p-2}{T - MT + 1}\right)$$

Proof The constraint that the sum equals one holds and is proved as above. As for the inequality constraints for positivity, we suppose that

$$T - MT + 1 > 0 \Rightarrow T(1-M) > -1 \overset{1-M<0}{\Rightarrow} T < \frac{1}{M-1}, \quad \text{Contradiction}$$

Hence the denominator is a negative quantity. For this point to represent fractions, we must therefore have:

$$\begin{aligned} &3p - 2 > 0 \ \ \& \ \ T + 3p - MT - 1 < 0 \\ &p > \frac{2}{3} \ \ \& \ \ T > \frac{3p-1}{M-1} \end{aligned}$$

For the second inequality to hold, the quantity on the right must be smaller than one, since by definition T is a natural number. It follows that:

$$3p - 1 < M - 1 \Rightarrow M > 3p \quad \text{where for this to hold for every p we must have } M \geq 3 \quad \blacksquare$$

Proposition 3.3.4 For every $M \in \{6, 7, \dots\}$ and every sufficiently large T, the following is an admissible equilibrium point.

$$\left(\frac{2(3T + 4p - MT + Mp - 3Tp - 3)}{2M + 6T - 3MT - 4}, 0, \frac{2M - 8p - MT - 2Mp + 6Tp + 2}{2M + 6T - 3MT - 4} \right) \tag{4}$$

Proof It holds that $x_1 + x_2 + x_3 = 1$. As for the denominator, suppose:

$$\begin{aligned} &2M + 6T - 3MT - 4 > 0 \Rightarrow 2M + T(6 - 3M) - 4 > 0 \overset{6-3M<0}{\Rightarrow} \\ &T < \frac{2M - 4}{3M - 6} \end{aligned}$$

$$2M - 4 < 3M - 6 \Rightarrow M > 2$$

which holds, so the inequality above leads to a contradiction, since T cannot be smaller than one. Hence, the denominator is always negative. To prove that the quantities of the point represent fractions, it must therefore be proven that

$$2(3T + 4p - MT + Mp - 3Tp - 3) < 0 \quad \& \quad 2M - 8p - MT - 2Mp + 6Tp + 2 < 0$$

For sufficiently large T, the sign of these expressions depends only on the term multiplied by T:

$$\begin{aligned} &3 - M - 3p < 0 \quad \& \quad 6p - M < 0 \\ &M > 3 - 3p \quad \& \quad M > 6p \end{aligned}$$

The first inequality holds for every p within the permitted range, while for the second inequality to hold, the constraint $M \geq 6$ must be introduced $\blacksquare$

Proposition 3.3.5 For every $M \in \{6, 7, \dots\}$ and every sufficiently large T, the following is an admissible equilibrium point.

$$\left[\begin{array}{c} -\frac{2(3p-2)(T+p+Mp-MT^2+T^2-2)}{M^2T^2+2M^2p^2-2M^2p-5MT^2-MT+4Mp^2-4Mp+4M+4T^2+4T+2p^2-2p-4}, \\ \frac{M^2T^2+2M^2p^2-2M^2p-6MT^2p-MT^2-6MTp^2+7MTp-3MT+10Mp^2-8Mp+4M+6T^2p-6Tp^2+16Tp-4T+8p^2-18p+4}{M^2T^2+2M^2p^2-2M^2p-5MT^2-MT+4Mp^2-4Mp+4M+4T^2+4T+2p^2-2p-4}, \\ \frac{4T+2MT-10Tp+6Tp^2-7MTp+6MTp^2}{M^2T^2+2M^2p^2-2M^2p-5MT^2-MT+4Mp^2-4Mp+4M+4T^2+4T+2p^2-2p-4} \end{array} \right]$$

Since the proof proceeds in a manner analogous to the above, it will not be given in detail due to the complexity of the expressions ■

The analysis of the stability of the admissible equilibrium points above follows. The eigenvalues of the Jacobian matrix of the differentials of equation (8) at each point were used. The Jacobian matrix is given by the following formulas for each of its elements.

- $J_{11} = 3Tx_2 - 3x_3 + 3Tx_3 + 4px_3 + 4x_1x_3 - x_2x_3 - Tx_2^2 + MTx_1 + Mpx_3 - 3Tpx_2 - 3Tpx_3 - 2Mx_1x_3 - 6Tx_1x_2 - 6Tx_1x_3 - 2Tx_2x_3 - \frac{3MTx_1^2}{2} - MTx_3^2$
- $J_{12} = -x_1\left(3Tx_1 + 2Tx_2 + 3T(p-1) + x_3(2T+1)\right)$
- $J_{13} = -x_1\left(x_1(M+3T-2) + (3T-3)(p-1) - p(M+1) + x_2(2T+1) + 2MTx_3\right)$
- $J_{21} = -x_2\left(3Tx_2 - 3Tp + x_3(M+3T-2) + MTx_1\right)$
- $J_{22} = 2Tx_2 - x_3 + Tx_3 + 3px_3 + 2x_1x_3 - 2x_2x_3 - 3Tx_2^2 + 3Tpx_1 - Mx_1x_3 - 6Tx_1x_2 - 3Tx_1x_3 - 4Tx_2x_3 - \frac{MTx_1^2}{2} - MTx_3^2$
- $J_{23} = -x_2\left(x_1(M+3T-2) - 3p - T + x_2(2T+1) + 2MTx_3 + 1\right)$
- $J_{31} = -x_3\left(3Tx_2 + x_3(M+3T-2) + (M+1)(p-1) - p(3T-3) + MTx_1\right)$
- $J_{32} = -x_3\left(3p - T + 3Tx_1 + 2Tx_2 + x_3(2T+1) - 2\right)$
- $J_{33} = x_1 + 2x_2 + Mx_1 + Tx_2 - 4px_1 - 3px_2 + 4x_1x_3 - 2x_2x_3 - Tx_2^2 + 2MTx_3 - Mpx_1 + 3Tpx_1 - 2Mx_1x_3 - 3Tx_1x_2 - 6Tx_1x_3 - 4Tx_2x_3 - \frac{MTx_1^2}{2} - 3MTx_3^2$

Proposition 3.3.6 Regarding the equilibrium point

$$(1, 0, 0)$$

For $M > 6p$ and for sufficiently large T, the equilibrium point is asymptotically stable.

Proof The Jacobian is:

$$J((1,0,0)) = \begin{bmatrix} -\frac{MT}{2} & -3Tp & 4p - M + Mp - 3Tp - 1 \\ 0 & -\frac{T(M-6p)}{2} & 0 \\ 0 & 0 & p(3T-3) - (M+1)(p-1) - \frac{MT}{2} \end{bmatrix}$$

Its eigenvalues are:

$$\lambda_1 = -\frac{MT}{2} < 0$$
$$\lambda_2 = -\frac{T(M-6p)}{2} < 0 \quad \text{iff} \quad M > 6p$$
$$\lambda_3 = p(3T-3) - (M+1)(p-1) - \frac{MT}{2} < 0$$

For λ_3, it holds that for sufficiently large T its sign is determined by the first-order terms in T, that is:

$$3p - \frac{M}{2} < 0 \Rightarrow M > 6p$$

Hence, for the point to be asymptotically stable, the above constraint must hold ■

Proposition 3.3.7 Regarding the equilibrium point

$$(0, 1, 0)$$

For $p > \frac{2}{3}$, and for sufficiently large T, the equilibrium point is asymptotically stable.

Proof The Jacobian is:

$$J((0,1,0)) = \begin{bmatrix} -T(3p-2) & 0 & 0 \\ 3T(p-1) & -T & 3p-T-2 \\ 0 & 0 & 2-3p \end{bmatrix}$$

Its eigenvalues are:

$$\begin{aligned} \lambda_1 &= 2-3p \\ \lambda_2 &= -T(3p-2) \\ \lambda_3 &= -T \end{aligned}$$

For $\lambda_1 < 0, \lambda_2 < 0$ to hold, the constraint $p > \frac{2}{3}$ must also hold. Moreover, λ_3 is always negative, so the equilibrium point under examination is asymptotically stable ■

Proposition 3.3.8 Regarding the equilibrium point

$$(0, 0, 1)$$

For every $M \in \{3, 4, \dots\}$ and for sufficiently large T, the equilibrium point is asymptotically stable.

Proof The Jacobian is:

$$J((0,0,1)) = \begin{bmatrix} p(M+1)-(3T-3)(p-1)-MT & 0 & 0 \\ 0 & T+3p-MT-1 & 0 \\ 3Tp-4p-Mp-3T+3 & 1-3p-T & -MT \end{bmatrix}$$

Its eigenvalues are:

$$\begin{aligned} \lambda_1 &= -MT < 0 \\ \lambda_2 &= p(M+1)-(3T-3)(p-1)-MT \\ \lambda_3 &= T+3p-MT-1 < 0 \quad \text{the proof was given in Proposition 3.3.3} \end{aligned}$$

Since the eigenvalues λ_1 and λ_3 are always negative quantities, it remains to determine the sign of the eigenvalue λ_2. For sufficiently large T its sign is determined by the first-order terms in T, that is:

$$T(-3p+3-M) < 0 \Rightarrow M > 3-3p$$

For the above inequality to hold, the constraint $M \in \{3, 4, \dots\}$ must also hold. Hence the equilibrium point under examination is also asymptotically stable for $M \in \{3, 4, \dots\}$ ■

Proposition 3.3.9 Regarding the equilibrium point

$$\left(\frac{6p-4}{M-4}, \frac{M-6p}{M-4}, 0\right)$$

For $M \in \{6, 7...\}$, $p \in (\frac{2}{3}, 1)$ and sufficiently large T, the equilibrium point is unstable.

Proof The eigenvalues of the Jacobian are:

$$\lambda_1 = \frac{(3p-2)(M-2p-2Mp+2)}{M-4}, \qquad \lambda_2 = -\frac{T(18p^2-18p+M)}{M-4}, \qquad \lambda_3 = \frac{T(3p-2)(M-6p)}{M-4}$$

Regarding the eigenvalue λ_1, the denominator is positive, and so is $(3p-2)$. Suppose that:

$$M-2p-2Mp+2>0 \Rightarrow M(1-2p)>2p-2 \overset{1-2p<0}{\Rightarrow} M<\frac{2-2p}{2p-1}$$

Contradiction, since $2-2p<2p-1 \Rightarrow p>\frac{1}{4}$ and M is a natural number. Hence the eigenvalue $\lambda_1<0$.

Regarding the eigenvalue λ_2, the denominator is positive and the sign of the numerator is proved to be positive as well; multiplied by $-T$, it is negative. Regarding the eigenvalue λ_3, $M-6p>0$, so it is positive. Finally, it follows that the equilibrium point under examination is unstable ■

Proposition 3.3.10 Regarding the equilibrium point

$$\left(0, \frac{T+3p-MT-1}{T-MT+1}, -\frac{3p-2}{T-MT+1}\right)$$

For $M \in \{3,4,\dots\}$, $p \in (\frac{2}{3},1)$ and sufficiently large T, it is unstable.

Proof The eigenvalues of the Jacobian are:

$$\lambda_1 = \frac{(3p-2)(T+3p-MT-1)}{T-MT+1}$$
$$\lambda_2 = -\frac{(3p-2)(T+p+Mp-MT^2+T^2-2)}{T-MT+1}$$
$$\lambda_3 = -\frac{T+9p-MT^2+T^2-9p^2-2}{T-MT+1}$$

Regarding the eigenvalue λ_1, both the denominator and the numerator are negative quantities, hence $\lambda_1>0$. Regarding the eigenvalue λ_2, the term $T+p+Mp-MT^2+T^2-2$ must be examined, where for sufficiently large T its sign is determined by the second-order terms in T, that is $T^2(1-M)<0$. Hence the eigenvalue $\lambda_2<0$. In the same way it is proved that $\lambda_3<0$ as well. However, since the eigenvalue $\lambda_1>0$, the point under examination is unstable ■

Proposition 3.3.11 Regarding the equilibrium point

$$\left(\frac{2(3T+4p-MT+Mp-3Tp-3)}{2M+6T-3MT-4}, 0, \frac{2M-8p-MT-2Mp+6Tp+2}{2M+6T-3MT-4}\right)$$

For $M \in \{6,7,\dots\}$ and sufficiently large T, it is unstable.

Proof The eigenvalues of the Jacobian for the above point have denominator $(3T-2)(M-2)$. Its sign turns out to be positive since

$$M>2 \quad \& \quad T>\frac{2}{3}$$

Since T is sufficiently large, in the numerator the term multiplied by T^2 dominates and determines the sign, so the following expressions result:

$$\lambda_1 = -\frac{T^2(M^2+18p^2-18p)}{(3T-2)(M-2)}, \quad \lambda_2 = -\frac{T^2(M^2-6Mp-M+6p)}{(3T-2)(M-2)}, \quad \lambda_3 = \frac{T^2(M^2-3Mp-3M-18p^2+18p)}{(3T-2)(M-2)}$$

It is proven that:

$$M^2+18p^2-18p>0 \quad \text{hence } \lambda_1<0$$
$$M^2-6Mp-M+6p=(M-1)(M-6p)>0 \quad \text{hence } \lambda_2<0$$
$$M^2-3Mp-3M-18p^2+18p>0 \quad \text{hence } \lambda_3>0$$

Based on the above inequalities, it is proved that the equilibrium point is unstable ■

Proposition 3.3.12 Regarding the equilibrium point

$$\left[\begin{array}{c} -\dfrac{2(3p-2)(T+p+Mp-MT^2+T^2-2)}{M^2T^2+2M^2p^2-2M^2p-5MT^2-MT+4Mp^2-4Mp+4M+4T^2+4T+2p^2-2p-4}, \\ \dfrac{M^2T^2+2M^2p^2-2M^2p-6MT^2p-MT^2-6MTp^2+7MTp-3MT+10Mp^2-8Mp+4M+6T^2p-6Tp^2+16Tp-4T+8p^2-18p+4}{M^2T^2+2M^2p^2-2M^2p-5MT^2-MT+4Mp^2-4Mp+4M+4T^2+4T+2p^2-2p-4}, \\ \dfrac{4T+2MT-10Tp+6Tp^2-7MTp+6MTp^2}{M^2T^2+2M^2p^2-2M^2p-5MT^2-MT+4Mp^2-4Mp+4M+4T^2+4T+2p^2-2p-4} \end{array}\right]$$

For $M \in \{6, 7, \dots\}$ and for sufficiently large T, the equilibrium point is unstable.

Since the proof proceeds in a manner analogous to the above, it will not be given in detail due to the complexity of the expressions ■

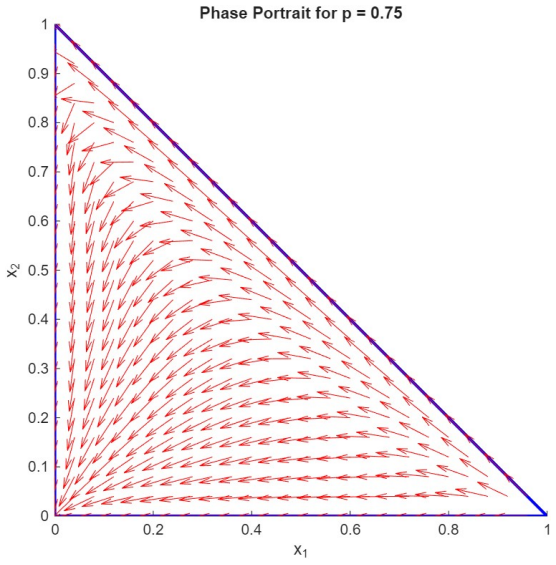


Figure 2: Phase portrait for $p = \frac{3}{4},\ M = 4,\ T = 10$.

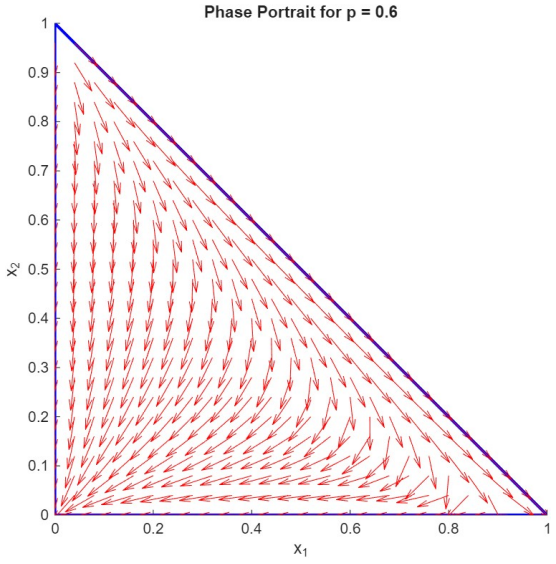
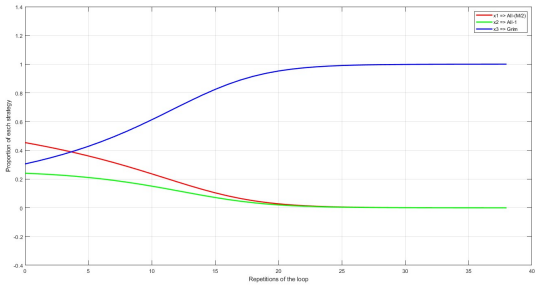
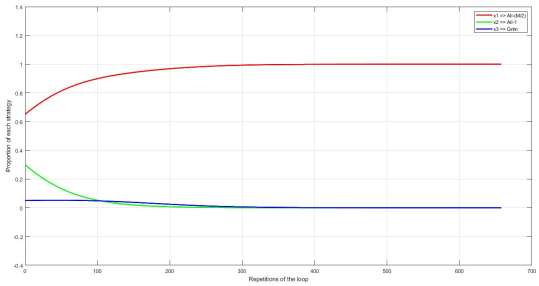


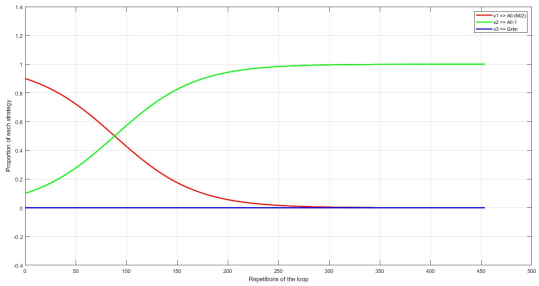
Figure 3: Phase portrait for $p = \frac{3}{5},\ M = 4,\ T = 10$.



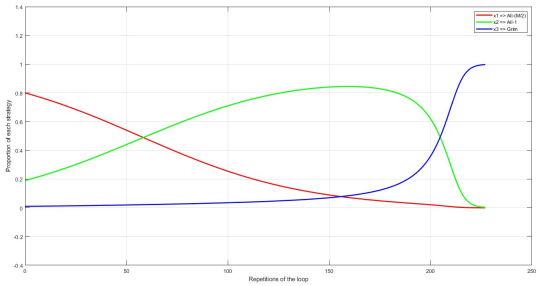
(a) $p = \frac{3}{4}$, random x_0.



(b) $p = \frac{3}{5}$, $x_0 = [0.65, 0.3, 0.05]$.



(c) $p = \frac{3}{4}$, $x_0 = [0.9, 0.1, 0]$.



(d) $p = \frac{3}{4}$, $x_0 = [0.8, 0.19, 0.01]$.

Figure 4: Results of replicator dynamics simulations for different parameter values and initial conditions.

It is observed that the fractions, following the arrows, tend to end up with the Grim strategy becoming established. However, if this strategy initially receives a zero fraction, then the game ends with the establishment of the strategy $All - 1$. For the majority of initial conditions the system tends to establish the Grim strategy. Here too there is a choice of initial fractions for which a different strategy becomes established, specifically the strategy $All - \frac{M}{2}$, but it is not necessary for the fraction following the Grim strategy to be zero. Based on the phase diagram, we expect that in most iterations the initial vector is driven to the equilibrium point $[0, 0, 1]$.

3.4 Evolutionarily Stable Strategies

Definition 3.4.1 A probability vector $\hat{x}$ is called an evolutionarily stable strategy (ESS) if and only if it satisfies the following conditions

$$\forall x : \hat{x}^T C \hat{x} \geq x^T C \hat{x} \tag{5}$$

$$\forall x \neq \hat{x} : \hat{x} C x > x^T C x \tag{6}$$

The following propositions characterize the symmetric Nash equilibrium points with respect to evolutionary stability.

Proposition 3.4.2 For $M < 6p$ ($M \geq 6p$), and sufficiently large T, the equilibrium point $(1, 0, 0)$ is not an ESS (is an ESS).

Proof In the case where $M < 6p$, the proposition is true, since for these values of p the equilibrium point is unstable. Hence the case $M \geq 6p$ remains. Let $\hat{x} = (1, 0, 0)$ and $x = (a, b, 1 - a - b)$ with $a, b, 1 - a - b \in (0, 1)$. Starting from condition (5) we have:

$$\hat{x}^T C \hat{x} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} \frac{TM}{2} & 3T(1-p) & (M+1)p + (T-1)3(1-p) \\ 3Tp & T & 3p + (T-1) \\ (M+1)(1-p) + (T-1)3p & 3(1-p) + (T-1) & TM \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \frac{TM}{2}$$

$$x^T C\hat{x} = \begin{bmatrix} a & b & 1-a-b \end{bmatrix} \begin{bmatrix} \frac{TM}{2} & 3T(1-p) & (M+1)p + (T-1)3(1-p) \\ 3Tp & T & 3p + (T-1) \\ (M+1)(1-p) + (T-1)3p & 3(1-p) + (T-1) & TM \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$= ((M+1)(p-1) - p(3T-3))(a+b-1) + \frac{aMT}{2} + 3Tbp$$

Consequently:

$$\hat{x}^T C\hat{x} - x^T C\hat{x} = T(1-a)(\frac{M}{2} - 3p) + (M - Mp - 1 + 4p)(1-a-b)$$

For sufficiently large T, the sign of the quantity $\hat{x}^T C\hat{x} - x^T C\hat{x}$ is determined by the first-order terms in T, that is:

$$T(1-a)(\frac{M}{2} - 3p) \geq 0$$

which holds for every $M \geq 6p$, and since condition (5) is satisfied, condition (6) does not need to be checked. Therefore $(0,1,0)$ is an ESS for $p \geq \frac{2}{3}$ ■

Proposition 3.4.3 For $p \geq \frac{2}{3}$ ($p < \frac{2}{3}$), $M, N > 1$ and sufficiently large T, the equilibrium point $(0,1,0)$ is an ESS (is not an ESS).

Proof In the case where $p < \frac{2}{3}$, the proposition is true, since for these values of p the equilibrium point is unstable. Hence the case $p \geq \frac{2}{3}$ remains. Let $\hat{x} = (0,1,0)$ and $x = (a, b, 1-a-b)$ with $a, b \in (0,1)$. Starting from condition (5) we have:

$$\hat{x}^T C\hat{x} = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{TM}{2} & 3T(1-p) & (M+1)p + (T-1)3(1-p) \\ 3Tp & T & 3p + (T-1) \\ (M+1)(1-p) + (T-1)3p & 3(1-p) + (T-1) & TM \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = T$$

$$x^T C\hat{x} = \begin{bmatrix} a & b & 1-a-b \end{bmatrix} \begin{bmatrix} \frac{TM}{2} & 3T(1-p) & (M+1)p + (T-1)3(1-p) \\ 3Tp & T & 3p + (T-1) \\ (M+1)(1-p) + (T-1)3p & 3(1-p) + (T-1) & TM \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$= Tb - (a+b-1)(T-3p+2) - 3Ta(p-1)$$

Consequently,

$$\hat{x}^T C\hat{x} - x^T C\hat{x} = Ta(3p-2) + (3p-2)(a+b-1)$$

For sufficiently large T, the sign of the quantity $\hat{x}^T C\hat{x} - x^T C\hat{x}$ is determined by the first-order terms in T, that is:

$$Ta(3p-2) \geq 0$$

which holds for every $M \geq 6p$, and since condition (5) is satisfied, condition (6) does not need to be checked. Therefore $(0,1,0)$ is an ESS for $M \geq 6p$ ■

Proposition 5.4 For every $M, N > 1$ and sufficiently large T, the equilibrium point $(0,0,1)$ is an ESS.

Proof Let $\hat{x} = (0,0,1)$ and $x = (a, b, 1-a-b)$ with $a, b \in (0,1)$. Starting from condition (5) we have:

$$\hat{x}^T C\hat{x} = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{TM}{2} & 3T(1-p) & (M+1)p + (T-1)3(1-p) \\ 3Tp & T & 3p + (T-1) \\ (M+1)(1-p) + (T-1)3p & 3(1-p) + (T-1) & TM \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = TM$$

$$x^T C\hat{x} = \begin{bmatrix} a & b & 1-a-b \end{bmatrix} \begin{bmatrix} \frac{TM}{2} & 3T(1-p) & (M+1)p+(T-1)3(1-p) \\ 3Tp & T & 3p+(T-1) \\ (M+1)(1-p)+(T-1)3p & 3(1-p)+(T-1) & TM \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$
$$= b(T+3p-1) - a[(3T-3)(p-1) - p(M+1)] - MT(a+b-1)$$

Consequently,

$$\hat{x}^T C\hat{x} - x^T C\hat{x} = MT - bT - 3bp + b + 3aTp - 3aT - 3ap + 3a - aMp - ap + MTa + MTb - MT$$

$$= T(-b + 3ap + Ma + Mb) + (-Mpa - 4ap - 3a - 3bp + b)$$

For sufficiently large T, the sign of the quantity $\hat{x}^T C\hat{x} - x^T C\hat{x}$ is determined by the first-order terms in T, that is:

$$T(-b + 3ap + Ma + Mb) \geq T(-b + Ma + Mb)$$
$$= T[Ma + (M-1)b]$$

Since $a, b \in (0,1)$ and $M \geq 1$ (otherwise the centipede game would be trivial), the first-order term in T will always be positive and condition (5) will be satisfied. Consequently, condition (6) does not need to be checked and we conclude that $(0,0,1)$ is indeed an ESS ■

Proposition 3.4.5 For suitable M, N, p, as mentioned previously in propositions (3.3.9), (3.3.10), (3.3.11) and (3.3.12) respectively, the equilibrium points

$$(\frac{6p-4}{M-4}, \frac{M-6p}{M-4}, 0)$$

$$(0, \frac{T+3p-MT-1}{T-MT+1}, -\frac{3p-2}{T-MT+1})$$

$$\left(\frac{2(3T+4p-MT+Mp-3Tp-3)}{2M+6T-3MT-4}, 0, \frac{2M-8p-MT-2Mp+6Tp+2}{2M+6T-3MT-4}\right)$$

$$\left[-\frac{2(3p-2)(T+p+Mp-MT^2+T^2-2)}{M^2T^2+2M^2p^2-2M^2p-5MT^2-MT+4Mp^2-4Mp+4M+4T^2+4T+2p^2-2p-4}, \right.$$
$$\frac{M^2T^2+2M^2p^2-2M^2p-6MT^2p-MT^2-6MTp^2+7MTp-3MT+10Mp^2-8Mp+4M+6T^2p-6Tp^2+16Tp-4T+8p^2-18p+4}{M^2T^2+2M^2p^2-2M^2p-5MT^2-MT+4Mp^2-4Mp+4M+4T^2+4T+2p^2-2p-4},$$
$$\left. \frac{4T+2MT-10Tp+6Tp^2-7MTp+6MTp^2}{M^2T^2+2M^2p^2-2M^2p-5MT^2-MT+4Mp^2-4Mp+4M+4T^2+4T+2p^2-2p-4} \right]$$

are not ESS.

Proof The above equilibrium points are unstable, therefore they cannot possibly be ESS ■

3.5 Markov Analysis

The Markov approach transfers the analysis from the continuous domain (where we had fractions x_i) to a finite population of N individuals. The evolution of the population is not described by differential equations, but by a Markov chain.

We start with a symmetric two-player game described by an $M \times M$ bimatrix (A, A^T). We assume the existence of N players and define the following:

1. **Population vector** A population vector

$$z = (z_1, \ldots, z_N)$$

belonging to

$$Z = \{(z_1, \ldots, z_N) : \forall n,\ z_n \in \{1, \ldots, M\}\}.$$

which denotes the strategy used by each player: $z_n = m$ means that the n-th player uses strategy m.

2. **State/strategy vector** A state vector

$$s = (s_1, \ldots, s_M) \in S_M$$

that shows how many players use each strategy, that is, s_m is the number of m-users. It takes values in the state space

$$S_M = \left\{ (s_1, \ldots, s_M) \in \{0, 1, \ldots, N\}^M \;\middle|\; \sum_{m=1}^{M} s_m = N \right\}.$$

The following subsets of S_M are also defined:

$$S_M^{(t)} = \{(1, 0, \ldots, 0),\ (0, 1, 0, \ldots, 0),\ \ldots,\ (0, \ldots, 0, 1)\}$$

$$S_M^{(i)} = \left\{ (s_1, \ldots, s_M) \in \{1, \ldots, N-1\}^M \;\middle|\; \sum_{m=1}^{M} s_m = N \right\}$$

$$S_M^{(f,m)} = \left\{ (s_1, \ldots, s_M) \in \{1, \ldots, N-1\}^M \;\middle|\; \sum_{k=1}^{M} s_k = N,\ s_m = 0 \right\}.$$

Obviously, the set

$$\left\{ S_M^{(t)},\ S_M^{(i)},\ S_M^{(f,1)}, \ldots, S_M^{(f,M)} \right\}$$

is a partition of S_M.

3. **Frequency vector** A frequency vector

$$x = (x_1, \ldots, x_M)$$

where for every $m \in \{1, \ldots, M\}$ we define

$$x_m = \frac{s_m}{N}.$$

Thus x_m shows the fraction of m-users. The vector x takes values in the simplex Δ_M.

For a given payoff matrix A and number of players N, the evolutionary game consists of the following procedure.

1. An initial distribution of strategies among the players is chosen:

$$z^{(0)} = \left(z_1^{(0)}, \ldots, z_N^{(0)}\right) \in Z.$$

2. For each $j \in \{0, 1, \ldots, J-1\}$ the following are repeated:

- The following are computed:

$$s^{(j)} = \left(s_1^{(j)}, \ldots, s_M^{(j)}\right) \in S_M \quad \text{and} \quad x^{(j)} = \left(x_1^{(j)}, \ldots, x_M^{(j)}\right),$$

  where for every m:

$$s_m^{(j)} = \left|\{z_n^{(j)} : z_n^{(j)} = m\}\right|, \qquad x_m^{(j)} = \frac{s_m^{(j)}}{\sum_{m=1}^{M} s_m^{(j)}}.$$

- All possible matches between the players are played, that is, all pairs (n_1, n_2) with $n_1, n_2 \in \{1, \ldots, N\}$ and $n_1 \neq n_2$. In each match, n_1 and n_2 play one round of the game A.
- The n-th player (for $n \in \{1, \ldots, N\}$) accumulates a total payoff

$$q_n\left(z^{(j)}\right) = \sum_{n' \neq n} A_{z_n^{(j)}\, z_{n'}^{(j)}}.$$

- The m-th strategy accumulates a total payoff

$$Q_m\left(s^{(j)}\right) = \sum_{n : z_n^{(j)} = m} q_n\left(z^{(j)}\right).$$

- The next population vector/generation is obtained

$$z^{(j+1)},$$

  which allows a player to update his strategy according to a revision protocol R.

The payoff matrix for the strategies is:

$$Q(s_1, s_2, s_3) = \begin{bmatrix} \frac{TM}{2}(s_1 - 1) + 3T(1-p)s_2 + [(M+1)p + (T-1)3(1-p)]s_3 \\ 3Tps_1 + T(s_2 - 1) + [3p + (T-1)]s_3 \\ [(M+1)(1-p) + (T-1)3p]s_1 + [3(1-p) + (T-1)]s_2 + TM(s_3 - 1) \end{bmatrix}$$

Let $R(s_1, s_2, s_3)$ be a 3×3 matrix for which

$$\forall i \neq j : R_{ij} = \max\left[0, Q_j(s_1, s_2, s_3) - Q_i(s_1, s_2, s_3)\right]$$

$$\forall i : R_{ii}(s_1, s_2, s_3) = \begin{cases} 0 & \text{iff } \max_{i \neq j}\left[Q_j(s_1, s_2, s_3)\right] > Q_i(s_1, s_2 s_3) \\ 1 & \text{otherwise} \end{cases}$$

Carrying out the computations without knowing the sign of the elements of the matrix R, the elements R_{ij} of the matrix are obtained as

$$R_{ij}(\mathbf{s}) = \bar{R}_{ij}(\mathbf{s}) + c_{ij}(\mathbf{s})$$

where the terms $c_{ij}(\mathbf{s})$ are constants and the terms $R_{ij}(\mathbf{s})$ are first-order in T. As in the preceding analyses, it is assumed that the sign of $R_{ij}(\mathbf{s})$ is determined by $\bar{R}_{ij}(\mathbf{s})$, and that $c_{ij}(\mathbf{s})$ is negligible. Consequently, there are six possible directions in which a state transition can occur, for which the corresponding $\bar{R}_{ij}(\mathbf{s})$ are obtained.

- Northwest $[(s_1, s_2, s_3) \to (s_1 - 1, s_2 + 1, s_3)]$

$$\bar{R}_{12}(s) = T\left((2 - \frac{M}{2})s_1 + N(3p - 2) - 1 + \frac{M}{2}\right)$$

- West $[(s_1, s_2, s_3) \to (s_1 - 1, s_2, s_3 + 1)]$

$$\bar{R}_{13}(s) = T\left((3 - \frac{3M}{2})s_1 - Ms_2 + N(3p - 3 + M) - \frac{M}{2}\right)$$

- Southeast $[(s_1, s_2, s_3) \to (s_1 + 1, s_2 - 1, s_3)]$

$$\bar{R}_{21}(s) = -T\left((2 - \frac{M}{2})s_1 + N(3p - 2) - 1 + \frac{M}{2}\right)$$

- South $[(s_1, s_2, s_3) \to (s_1, s_2 - 1, s_3 + 1)]$

$$\bar{R}_{23}(s) = T\left((1-M)s_1 + (1-N)s_2 + (N-1)(M-1)\right)$$

- East $[(s_1, s_2, s_3) \to (s_1 + 1, s_2, s_3 - 1)]$

$$\bar{R}_{31}(s) = -T\left((3 - \frac{3M}{2})s_1 - Ms_2 + N(3p - 3 + M) - \frac{M}{2}\right)$$

- North $[(s_1, s_2, s_3) \to (s_1, s_2 + 1, s_3 - 1)]$

$$\bar{R}_{32}(s) = -T\left((1-M)s_1 + (1-N)s_2 + (N-1)(M-1)\right)$$

Obviously, not all transitions are possible every time, since if, for example, for specific parameters and states it holds that $\bar{R}_{12} > 0$, then obviously $\bar{R}_{21} = 0$. Nevertheless, we keep analytic expressions for both transitions so that the conditions under which each transition is possible can be examined. Also, in the following propositions we exclude the self-transitions, since they do not affect the allowed transitions of a state (s_1, s_2, s_3).

Definition 3.5.1 The revision protocol Pairwise Proportional Imitation (PPI) is defined as follows:

1. A player is chosen from the population with equal probability.
2. If the player uses strategy m_1, then:
   (a) If
   $$Q_{m_1}(s) = \max_m Q_m(s),$$
   then the player keeps the strategy m_1 in the next iteration.
   (b) Otherwise, the player adopts strategy m_2 with probability
   $$\rho_{m_1 m_2}(s),$$
   where
   $$\rho_{m_1 m_2}(s) = \frac{x_{m_2}\left[Q_{m_2}(s) - Q_{m_1}(s)\right]}{\sum_{m=1}^{M} x_m \left[Q_m(s) - Q_{m_1}(s)\right]^+}.$$

In other words, the probability $\rho_{m_1 m_2}(s)$ is proportional to the positive part of the difference (surplus) of the payoff Q_{m_2} over Q_{m_1}, multiplied by the frequency of occurrence of strategy m_2 in the population.

Proposition 3.5.2 Let $M, N \in \{4, 5, \dots\}$ with $M \geq N$ and T sufficiently large. For every $(s_1, s_2, s_3) \in S_3^{(i)}$, excluding self-transitions, the only transitions of positive probability are the following:

south: $(s_1, s_2, s_3) \to (s_1, s_2 - 1, s_3 + 1)$

northwest: $(s_1, s_2, s_3) \to (s_1 - 1, s_2 + 1, s_3)$ iff $s_1 < \dfrac{\frac{M}{2} - 1 + N(3p-2)}{\frac{M}{2} - 2}$

southeast: $(s_1, s_2, s_3) \to (s_1 + 1, s_2 - 1, s_3)$ iff $s_1 > \dfrac{\frac{M}{2} - 1 + N(3p-2)}{\frac{M}{2} - 2}$

west: $(s_1, s_2, s_3) \to (s_1 - 1, s_2, s_3 + 1)$
iff $(3 - \frac{3M}{2})s_1 - Ms_2 + N(3p - 3 + M) - \frac{M}{2} > 0$

east: $(s_1, s_2, s_3) \to (s_1 + 1, s_2, s_3 - 1)$
iff $(3 - \frac{3M}{2})s_1 - Ms_2 + N(3p - 3 + M) - \frac{M}{2} < 0$

Proof The basic observation is that, for interior states (s_1, s_2, s_3), the PPI rule dictates that the transitions with positive probability can be characterized as follows:

$$\forall (s_1, s_2, s_3) \in S_3^{(i)} : P_{(s_1,s_2,s_3),(s_1',s_2',s_3')} > 0 \iff R_{ij}(s_1, s_2, s_3) > 0 \iff \rho_{ij}(s_1, s_2, s_3) > 0$$

Therefore, the above relations for the elements of the transition matrix R completely determine the transitions of positive probability. As an example, we give the transitions of the form

$$(s_1, s_2, s_3) \to (s_1 - 1, s_2 + 1, s_3)$$

and

$$(s_1, s_2, s_3) \to (s_1 + 1, s_2 - 1, s_3)$$

(that is, the "northwest" and "southeast" transitions). For the transition

$$(s_1, s_2, s_3) \to (s_1 - 1, s_2 + 1, s_3),$$

a player of type 1 must be chosen (which happens with probability $\frac{1}{3}$) and in addition it must hold that

$$R_{12}(s_1, s_2, s_3) > 0.$$

We know that

$$R_{12}(s_1, s_2, s_3) = T\Big((2 - \frac{M}{2})s_1 + N(3p - 2) - 1 + \frac{M}{2}\Big).$$

Hence

$$\Pr\Big((s_1, s_2, s_3) \to (s_1 - 1, s_2 + 1, s_3)\Big) > 0$$

if and only if

$$(2 - \frac{M}{2})s_1 + N(3p - 2) - 1 + \frac{M}{2} > 0.$$

Now let us apply the same reasoning, separating the relations for the elements of the transition matrix R into three groups.

1. For the elements $\bar{R}_{23}$ and $\bar{R}_{32}$ we have:

$$\bar{R}_{23} > 0 \iff (1 - M)s_1 + (1 - N)s_2 + (N - 1)(M - 1) > 0$$
$$\bar{R}_{32} > 0 \iff (1 - M)s_1 + (1 - N)s_2 + (N - 1)(M - 1) < 0$$

   The line

$$l_2 : (1 - M)s_1 + (1 - N)s_2 + (N - 1)(M - 1) = 0$$

   passes through the points $(0, M - 1)$ and $(N - 1, 0)$. In the case where $M \geq N$, the whole triangle of interior feasible points will lie in one half-plane, and since

$$(1 - M)0 + (1 - N)0 + (N - 1)(M - 1) = (N - 1)(M - 1) > 0$$

   this will be the positive half-plane. Hence it will always hold that $\bar{R}_{23} > 0$ and $\bar{R}_{32} < 0$, so we conclude that:

$$Pr((s_1, s_2, s_3) \to (s_1, s_2 - 1, s_3 + 1)) > 0$$
$$Pr((s_1, s_2, s_3) \to (s_1, s_2 + 1, s_3 - 1)) = 0$$

   that is, only the south transitions have positive probability.

2. For the elements $\bar{R}_{12}$ and $\bar{R}_{21}$ we have:

$$\bar{R}_{12} > 0 \iff (2 - \frac{M}{2})s_1 + N(3p-2) - 1 + \frac{M}{2} > 0 \iff s_1 < \frac{\frac{M}{2} - 1 + N(3p-2)}{\frac{M}{2} - 2}$$
$$\bar{R}_{21} > 0 \iff (2 - \frac{M}{2})s_1 + N(3p-2) - 1 + \frac{M}{2} < 0 \iff s_1 > \frac{\frac{M}{2} - 1 + N(3p-2)}{\frac{M}{2} - 2}$$

Hence, based on the above relations, it follows that:

$$s_1 < \frac{\frac{M}{2} - 1 + N(3p-2)}{\frac{M}{2} - 2} \iff Pr((s_1, s_2, s_3) \to (s_1 - 1, s_2 + 1, s_3)) > 0 \text{ (northwest)}$$
$$s_1 > \frac{\frac{M}{2} - 1 + N(3p-2)}{\frac{M}{2} - 2} \iff Pr((s_1, s_2, s_3) \to (s_1 + 1, s_2 - 1, s_3)) > 0 \text{ (southeast)}$$

3. For the elements $\bar{R}_{13}$ and $\bar{R}_{31}$ we have:

$$\bar{R}_{13} > 0 \iff (3 - \frac{3M}{2})s_1 - Ms_2 + N(3p - 3 + M) - \frac{M}{2} > 0$$
$$\bar{R}_{31} > 0 \iff -(3 - \frac{3M}{2})s_1 - Ms_2 + N(3p - 3 + M) - \frac{M}{2} < 0$$

The line

$$l_3 : (3 - \frac{3M}{2})s_1 - Ms_2 + N(3p - 3 + M) - \frac{M}{2} = 0$$

passes through the points $(0, \frac{N(3p-3+M)-\frac{M}{2}}{M})$ and $(\frac{N(3p-3+M)-\frac{M}{2}}{3-\frac{3M}{2}}, 0)$ and splits the triangle into two half-planes, so both transitions are feasible and occur depending on the choice of parameters.

Finally we have that:

$$(3 - \frac{3M}{2})s_1 - Ms_2 + N(3p - 3 + M) - \frac{M}{2} > 0 \iff Pr((s_1, s_2, s_3) \to (s_1 - 1, s_2, s_3 + 1)) > 0 \text{ (west)}$$
$$(3 - \frac{3M}{2})s_1 - Ms_2 + N(3p - 3 + M) - \frac{M}{2} < 0 \iff Pr((s_1, s_2, s_3) \to (s_1 + 1, s_2, s_3 - 1)) > 0 \text{ (east)} \blacksquare$$

Proposition 6.2 Under the assumption that $M, N \in \{4, ...\}$ and T sufficiently large, for any state $(0, s_2, s_3) \in S_3^{(1)}$, excluding self-transitions, the only feasible transitions are the following:

north: when in the initial state $s_2 > M - 1 : (0, s_2, s_3) \to (0, s_2 + 1, s_3 - 1)$
south: when in the initial state $s_2 < M - 1 : (0, s_2, s_3) \to (0, s_2 - 1, s_3 + 1)$

Proof The only transitions that need to be examined are $\bar{R}_{23}$ and $\bar{R}_{32}$, so setting s_1 to zero we obtain:

$$\bar{R}_{23} = T\left((1 - N)s_2 + (1 - N)(1 - M)\right)$$
$$\bar{R}_{32} = -T\left((1 - N)s_2 + (1 - N)(1 - M)\right)$$

Examining the transition $\bar{R}_{23}$ we have:

$$(1 - N)s_2 + (1 - N)(1 - M) > 0 \Rightarrow (1 - N)(s_2 + 1 - M) > 0 \overset{1-N<0}{\Rightarrow} s_2 + 1 - M < 0 \Rightarrow s_2 < M - 1$$
$$(1 - N)s_2 + (1 - N)(1 - M) < 0 \Rightarrow ... \Rightarrow s_2 > M - 1$$

(Based on the Markov diagrams, this proposition is not verified)

Proposition 3.5.3 Under the assumption that $M, N \in \{4, ...\}$ and T sufficiently large, for any state $(s_1, 0, s_3) \in S_3^{(2)}$, excluding self-transitions, the only feasible transitions are the following:

$$\text{east : when in the initial state } s_1 > \frac{\frac{M}{2} - N(3p-3+M)}{3-\frac{3M}{2}} : (s_1, 0, s_2) \to (s_1+1, 0, s_3-1)$$
$$\text{west : when in the initial state } s_1 < \frac{\frac{M}{2} - N(3p-3+M)}{3-\frac{3M}{2}} : (s_1, 0, s_2) \to (s_1-1, 0, s_3+1)$$

Proof The only transitions that need to be examined are $\bar{R}_{13}$ and $\bar{R}_{31}$, so setting s_2 to zero we obtain:

$$\bar{R}_{13} = T\left((3-\frac{3M}{2})s_1 + N(3p-3+M) - \frac{M}{2}\right)$$
$$\bar{R}_{31} = -T\left((3-\frac{3M}{2})s_1 + N(3p-3+M) - \frac{M}{2}\right)$$

Examining the transition $\bar{R}_{13}$ we have:

$$(3-\frac{3M}{2})s_1 + N(3p-3+M) - \frac{M}{2} > 0 \overset{3-\frac{3M}{2}<0}{\Rightarrow} s_1 < \frac{\frac{M}{2} - N(3p-3+M)}{3-\frac{3M}{2}}$$
$$(3-\frac{3M}{2})s_1 + N(3p-3+M) - \frac{M}{2} < 0 \overset{3-\frac{3M}{2}<0}{\Rightarrow} s_1 > \frac{\frac{M}{2} - N(3p-3+M)}{3-\frac{3M}{2}}$$

This completes the proof ■

Proposition 3.5.4 Under the assumption that $M, N \in \{4, ...\}$ and T sufficiently large, for any state $(s_1, s_2, 0) \in S_3^{(3)}$, excluding self-transitions, the only feasible transitions are the following:

$$\text{northwest: when } (3p-\frac{M}{2})s_1 + (3p-2)s_2 + (\frac{M}{2}-1) < 0 : (s_1, s_2, 0) \to (s_1+1, s_2-1, 0)$$
$$\text{southeast: when } (3p-\frac{M}{2})s_1 + (3p-2)s_2 + (\frac{M}{2}-1) > 0 : (s_1, s_2, 0) \to (s_1-1, s_2+1, 0)$$

Proof The only transitions that need to be examined are of the form:

$$(s_1, s_2, 0) \to (s_1+1, s_2-1, 0)$$
$$(s_1, s_2, 0) \to (s_1-1, s_2+1, 0)$$

This holds since any transition involving s_3 is impossible, as the population of the Grim strategy is zero. Modifying the formulas for the transitions $\bar{R}_{12}, \bar{R}_{21}$ so that s_3 appears, we have:

$$\bar{R}_{12} = T\left(\left(3p-\frac{M}{2}\right)s_1 + (3p-2)\,s_2 + (3p-2)s_3 + \left(\frac{M}{2}-1\right)\right)$$
$$\bar{R}_{21} = -T\left(\left(3p-\frac{M}{2}\right)s_1 + (3p-2)s_2 + (3p-2)s_3 + \left(\frac{M}{2}-1\right)\right)$$

Setting s_3 to zero, the sign of the transitions is finally determined by the expression:

$$\left(3p-\frac{M}{2}\right)s_1 + (3p-2)\,s_2 + \left(\frac{M}{2}-1\right)$$

Hence we have:

$$(3p-\frac{M}{2})s_1 + (3p-2)s_2 + (\frac{M}{2}-1) > 0 \Rightarrow Pr((s_1, s_2, 0) \to (s_1-1, s_2+1, 0)) > 0$$
$$(3p-\frac{M}{2})s_1 + (3p-2)s_2 + (\frac{M}{2}-1) < 0 \Rightarrow Pr((s_1, s_2, 0) \to (s_1+1, s_2-1, 0)) > 0$$

This completes the proof ■

If we wanted to continue the analysis in order to arrive at more specific conditions for determining the sign of the transition relations, we could observe, through the Markov diagrams, that for $p < \frac{2}{3}$ the transitions of the form examined above are northwest, and for $p \geq \frac{2}{3}$ the transitions are southeast. Obviously, things are not so simple, because the choices of the remaining parameters also play a role.

Proposition 3.5.5 For the nodes $(N, 0, 0), (0, N, 0), (0, 0, N)$, the only transitions with non-zero probability are the transitions to themselves (self-transitions).

Proof For brevity, not all three vertices of the polyhedron will be studied, but only one of them, since the philosophy of the proof is the same. Regarding the node $(s_1, s_2, s_3) = (0, 0, N)$, the only possible transitions are north, east and to itself. For $N, M > 1$, it holds that

$$\bar{R}_{32}(0, 0, N) = -T\left[(1 - M)0 + (1 - M)0 + (M - 1)(N - 1)\right] = -T(M - 1)(N - 1) < 0$$

therefore the probability of a transition to the north is zero. Similarly

$$\bar{R}_{31}(0, 0, N) = -T\left[\left(3 - \frac{3M}{2}\right)0 + (1 - M)0 + N(M + 3p - 3) - \frac{M}{2}\right] = -T\left[N\left(M + 3p - 3\right) - \frac{M}{2}\right] < 0, \forall M, N \in 2, ...$$

Therefore, the node $(0, 0, N)$ will transition to itself with probability 1 ■

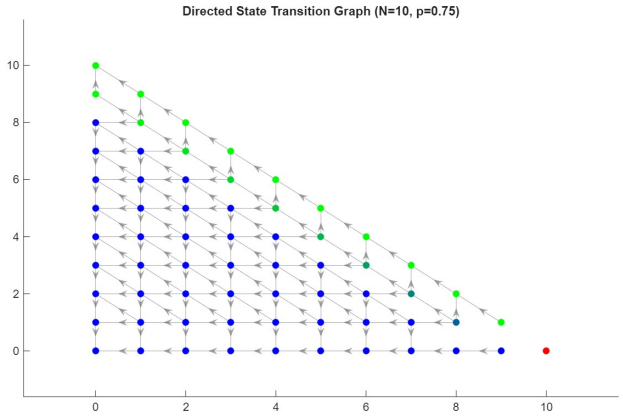


Figure 5: State transition graph for $p = \frac{3}{4}$, $M = 4$, $T = 10$ and $N = 10$.

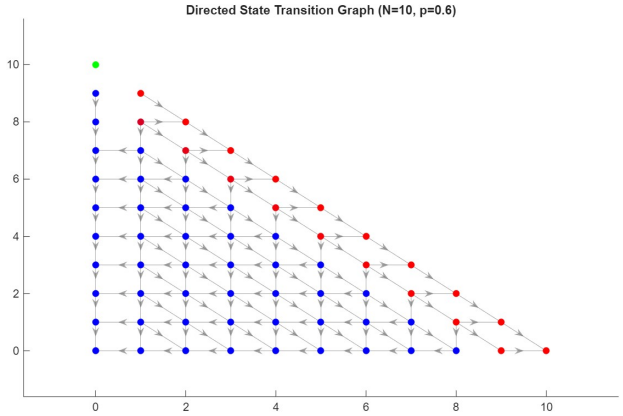
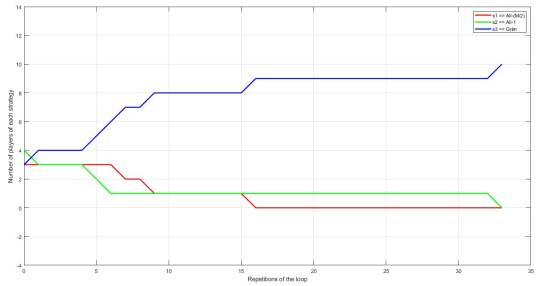
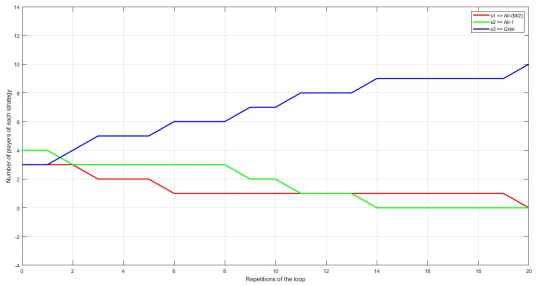


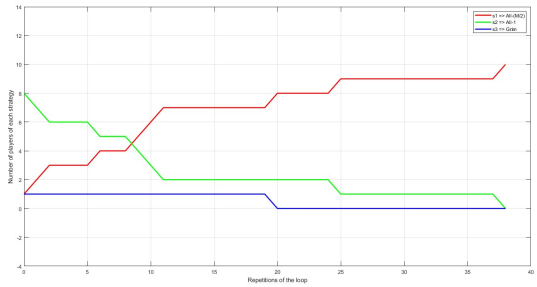
Figure 6: State transition graph for $p = \frac{3}{5}$, $M = 4$, $T = 10$ and $N = 10$.



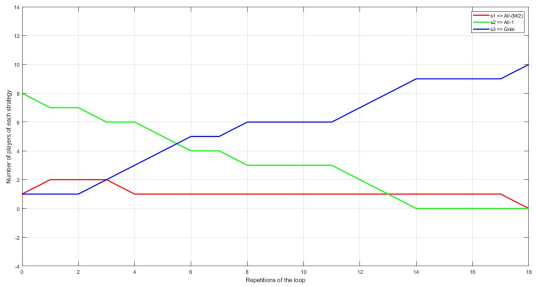
(a) This particular initial state was chosen because the state transition diagram shows that the game will end up in the equilibrium state $[0, 0, 10]$.



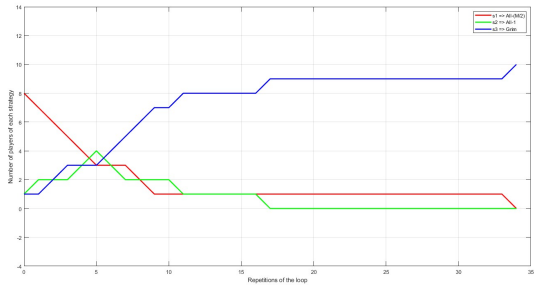
(b) This particular initial state was chosen because the state transition diagram shows that the game will end up in the equilibrium state $[0, 0, 10]$.



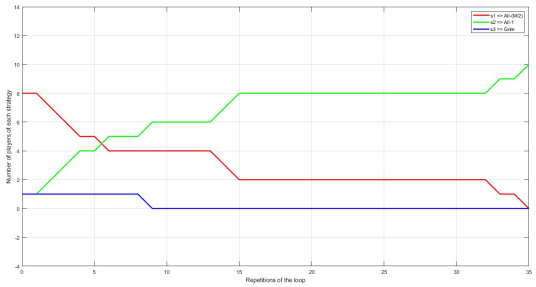
(c) The state transition diagram shows that there is a high probability (due to the color coding) that the initial state s_0 will be driven to the equilibrium state $[10, 0, 0]$.



(d) However, there is a probability that it will also be driven to the equilibrium state $[0, 0, 10]$. Since the program follows this probabilistic model, it does not give the same result in every iteration.



(e) Based on the color coding, the state transition diagram shows that with higher probability the initial state will be driven to the equilibrium state $[0, 0, 10]$.



(f) However, it is also possible for the game to end up in the equilibrium state $[0, 10, 0]$.

Figure 7: Results of Markov dynamics simulations for different values of the parameter p and different initial states s_0, with constant population size $N = 10$.

4 Implementation and Analysis with 5 Strategies

Following the same approach as with three strategies, the study of the evolutionary iterated game can be extended to five strategies.

4.1 Strategies and Payoffs

The first three are those analyzed in the thesis of Lazaridis-Kechagias

- $_1$ **or All-M**: in every OSC the strategy ρ_M is played
- $_2$ **or All-1**: as defined in (3.1)
- $_3$ **or Grim**: as defined in (3.1)

Some examples follow that explain the values $C_{11}, C_{12}, C_{13}, C_{21}, C_{22}, C_{23}, C_{31}, C_{32}, C_{33}$

1. The element C_{11} corresponds to the payoff of a player who uses the strategy $\sigma_{\text{All-M}}$ when playing against another player who also uses the strategy $\sigma_{\text{All-M}}$.

   Since in all T rounds the strategy ρ_M is played against ρ_M, the total payoff of the player is

   $$C_{11} = TA_{MM}.$$

2. Similarly, the element C_{12} corresponds to the payoff of a player who uses the strategy $\sigma_{\text{All-M}}$ when playing against a player who uses the strategy $\sigma_{\text{All-1}}$.

   Since in all T rounds the strategy ρ_M is played against ρ_1, the total payoff is

   $$C_{12} = TA_{M1}.$$

3. We now examine the element C_{23}, which corresponds to the payoff of a player who uses the strategy σ_G when playing against a player who uses the strategy $\sigma_{\text{All-1}}$.

   In the first round the σ_G player plays the strategy ρ_M against ρ_1 and receives payoff A_{M1}.

   In the remaining $T-1$ rounds the punishment mechanism of the Grim strategy is activated, so ρ_1 is played against ρ_1, yielding a total payoff of

   $$(T-1)A_{11}.$$

   Therefore, the total payoff is

   $$C_{23} = A_{M1} + (T-1)A_{11}.$$

The other two strategies were left as a free choice, and the ones chosen are[1]:

- σ_4 **or Old Testament**: The player who adopts this strategy plays ρ_M in the first round and in every round after it, provided that the opponent also plays ρ_M. If the opponent abandons the strategy ρ_M, then in the next round the player plays $\rho_{\frac{M}{2}}$. If the opponent plays something less than $\rho_{\frac{M}{2}}$, then the player plays $\rho_{\frac{M}{4}}$, and so on, until reaching the strategy ρ_1. This presupposes that M can be expressed as $M = 2^k$, with $k = 2, 3, ...$
- σ_5 **or New Testament**: The player who adopts this strategy plays ρ_M in the first round and in every round after it, provided that the opponent also plays ρ_M. If the opponent abandons the strategy ρ_M in the n-th round, then in the $(n+1)$-th round the player plays ρ_1, and in the $(n+2)$-th round plays ρ_M, regardless of what the opponent plays in the $(n+1)$-th round. This is repeated every time the player is "betrayed" by the opponent, that is, every time the opponent abandons the strategy ρ_M.

Using 5 strategies, the new matrix C has dimension 5×5, with the elements $C_{11}, C_{12}, C_{13}, C_{21}, ...C_{32}, C_{33}$, as well as the matrix A, remaining as computed in the thesis of Lazaridis-Kechagias. Some indicative examples of the remaining elements of the matrix follow.

1. We consider the case where the row player plays σ_1 and the column player plays σ_4. Since the row player never betrays the column player, the latter plays ρ_M in every round, and the payoff C_{14} is equal to the payoff C_{11}, as with all cases where the game is repeated in a fully cooperative manner, hence

   $$C_{14} = A_{MM} = TM$$

[1]The names of the two new strategies are based on the two different archetypes of God in the Bible. In the Old Testament He is described as strict and punishing, while in the New Testament He is lenient and forgiving.

2. We consider the case where the row player plays σ_1 and the column player plays σ_5. As before, since there is no "betrayal", the game is repeated in a fully cooperative manner, hence

$$C_{15} = A_{MM} = TM$$

3. We consider the case where the row player plays σ_2 and the column player plays σ_4. In the first round, the row player plays ρ_1 and the column player plays ρ_M, so the payoff of the first round is A_{1M}. In the next round, the row player plays ρ_1 and the column player plays $\rho_{\frac{M}{2}}$, so the payoff of the second round is $A_{1\frac{M}{2}}$. This process is repeated and the column player halves his threshold until it reaches one, that is, in the $(\log_2 M)$-th round. For the remaining $T - \log_2 M$ rounds, the payoff is A_{11}. Consequently,

$$C_{24} = \sum_{i=1}^{\log_2 M} A_{1\frac{M}{2^{i-1}}} + (T - \log_2 M)A_{11} = \sum_{i=1}^{\log_2 M} 3p + (T - \log_2 M) = 3p \log_2 M + (T - \log_2 M)$$

4. We consider the case where the row player plays σ_2 and the column player plays σ_5. In the first round the row player plays ρ_1 and the column player plays ρ_1, so the payoff of the round is A_{1M}. In the second round both players play ρ_1, so the payoff is A_{11}. This pattern is repeated for all T rounds, that is, in half of them on average the payoff is A_{1M} and in the other half A_{11}, therefore

$$C_{25} = \frac{T}{2}(A_{1M} + A_{11}) = \frac{T}{2}(1 + 3p)$$

All the elements of the matrix C are computed in a similar way, and it is now

$$C = \begin{bmatrix} TM & 3T(1-p) & TM & TM & TM \\ 3pT & T & 3p + (T-1) & 3p\log_2 M + (T - \log_2 M) & \frac{T}{2}(1+3p) \\ TM & 3(1-p) + (T-1) & TM & TM & TM \\ TM & 3(1-p)\log_2 M + (T - \log_2 M) & TM & TM & TM \\ TM & \frac{T}{2}(4-3p) & TM & TM & TM \end{bmatrix}$$

4.2 Equilibrium Points

The following propositions present the Nash equilibrium points of the new game with the five strategies.

Proposition 4.2.1 For $M \in \{4, 8, 16, ...\}$, the following is a symmetric NE.

$$((1,0,0,0,0),(1,0,0,0,0))$$

Proof As in the previous proofs, the sign of the payoff difference will be studied based on definition (1)

$Q_1((1,0,0,0,0),(1,0,0,0,0)) =$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} TM & 3T(1-p) & TM & TM & TM \\ 3pT & T & 3p + (T-1) & 3p\log_2 M + (T - \log_2 M) & \frac{T}{2}(1+3p) \\ TM & 3(1-p) + (T-1) & TM & TM & TM \\ TM & 3(1-p)\log_2 M + (T - \log_2 M) & TM & TM & TM \\ TM & \frac{T}{2}(4-3p) & TM & TM & TM \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} =$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} TM \\ 3pT \\ TM \\ TM \\ TM \end{bmatrix} = TM$$

Suppose that P_1 switches to the strategy $\tau = (\alpha, 1 - \alpha - b - c - d, b, c, d)$

$Q_1(\tau,(1,0,0,0,0)) =$

$$= \begin{bmatrix}\alpha & 1-\alpha-b-c-d & b & c & d\end{bmatrix}\begin{bmatrix}TM\\3pT\\TM\\TM\\TM\end{bmatrix} =$$

$$= \alpha TM + (1-\alpha-b-c-d)3pT + bTM + cTM + dTM$$

$\delta Q = Q_1((1,0,0,0,0),(1,0,0,0,0)) - Q_1(\tau,(1,0,0,0,0)) =$

$$= TM - \alpha TM - (1-\alpha-b-c-d)3pT - bTM - cTM - dTM$$
$$= TM(1-\alpha-b-c-d) - (1-\alpha-b-c-d)3pT$$
$$= T(1-\alpha-b-c-d)(M-3p)$$

For $M \in \{3,4,...\}$ the sign of δQ is always positive since $M > 3p$ and $1-\alpha-b-c-d > 0$, hence $((1,0,0,0,0),(1,0,0,0,0))$ is a NE ■

Proposition 4.2.2 For $M \in \{4,8,16,...\}$ and for $p > \frac{2}{3}$, the following is a symmetric NE.

$$((0,1,0,0,0),(0,1,0,0,0))$$

Proof As before,

$Q_1((0,1,0,0,0),(0,1,0,0,0)) =$

$$\begin{bmatrix}0 & 1 & 0 & 0 & 0\end{bmatrix}\begin{bmatrix}TM & 3T(1-p) & TM & TM & TM\\ 3pT & T & 3p+(T-1) & 3p\log_2 M + (T-\log_2 M) & \frac{T}{2}(1+3p)\\ TM & 3(1-p)+(T-1) & TM & TM & TM\\ TM & 3(1-p)\log_2 M + (T-\log_2 M) & TM & TM & TM\\ TM & \frac{T}{2}(4-3p) & TM & TM & TM\end{bmatrix}\begin{bmatrix}0\\1\\0\\0\\0\end{bmatrix} =$$

$$= \begin{bmatrix}0 & 1 & 0 & 0 & 0\end{bmatrix}\begin{bmatrix}3T(1-p)\\T\\3(1-p)+(T-1)\\3(1-p)log_2 M + (T-log_2 M)\\\frac{T}{2}(4-3p)\end{bmatrix} = T$$

Suppose that P_1 switches to the strategy $\tau = (\alpha, 1-\alpha-b-c-d, b, c, d)$

$Q_1(\tau,(0,1,0,0,0)) =$

$$= \begin{bmatrix}\alpha & 1-\alpha-b-c-d & b & c & d\end{bmatrix}\begin{bmatrix}3T(1-p)\\T\\3(1-p)+(T-1)\\3(1-p)log_2 M + (T-log_2 M)\\\frac{T}{2}(4-3p)\end{bmatrix} =$$

$Q_1 = \alpha[3T(1-p)] + (1-\alpha-b-c-d)T + b[3(1-p)+T-1] + c[3(1-p)\log_2 M + T - \log_2 M] + d\left[\frac{T}{2}(4-3p)\right]$

$Q_1(\tau,(0,1,0,0,0)) = T + (2-3p)[T(\alpha+0.5d) + b + c\log_2 M]$

Therefore we have

$$\delta Q = Q_1((1,0,0,0,0),(1,0,0,0,0)) - Q_1(\tau,(1,0,0,0,0)) = (3p-2)[T(\alpha+0.5d) + b + c\log_2 M]$$

The term inside the brackets is always positive by definition, so the sign of δQ is determined by the term $(3p-2)$; specifically, if $p > \frac{2}{3}$, then $\delta Q > 0$, and $((0,1,0,0,0),(0,1,0,0,0))$ is a NE ■

Proposition 4.2.3 For sufficiently large T and for $M \in \{3,4,...\}$, the following is a symmetric NE.

$$((0,0,1,0,0),(0,0,1,0,0))$$

Proof As before,

$$Q_1((0,0,1,0,0),(0,0,1,0,0)) =$$

$$\begin{bmatrix}0 & 0 & 1 & 0 & 0\end{bmatrix}\begin{bmatrix}TM & 3T(1-p) & TM & TM & TM\\ 3pT & T & 3p+(T-1) & 3p\log_2 M+(T-\log_2 M) & \frac{T}{2}(1+3p)\\ TM & 3(1-p)+(T-1) & TM & TM & TM\\ TM & 3(1-p)\log_2 M+(T-\log_2 M) & TM & TM & TM\\ TM & \frac{T}{2}(4-3p) & TM & TM & TM\end{bmatrix}\begin{bmatrix}0\\0\\1\\0\\0\end{bmatrix} =$$

$$= \begin{bmatrix}0 & 0 & 1 & 0 & 0\end{bmatrix}\begin{bmatrix}TM\\ 3p+(T-1)\\ TM\\ TM\\ TM\end{bmatrix} = TM$$

Suppose that P_1 switches to the strategy $\tau = (\alpha, 1-\alpha-b-c-d, b, c, d)$

$$Q_1(\tau,(0,0,1,0,0)) =$$

$$= \begin{bmatrix}\alpha & 1-\alpha-b-c-d & b & c & d\end{bmatrix}\begin{bmatrix}TM\\ 3p+(T-1)\\ TM\\ TM\\ TM\end{bmatrix} =$$

$$= \alpha TM + (1-\alpha-b-c-d)(3p+(T-1)) + bTM + cTM + dTM$$

$$\delta Q = Q_1((0,0,1,0,0),(0,0,1,0,0)) - Q_1(\tau,(0,0,1,0,0)) =$$

$$= TM - \alpha TM - (1-\alpha-b-c-d)(3p+(T-1)) - bTM - cTM - dTM$$

$$= TM(1-\alpha-b-c-d) - (1-\alpha-b-c-d)(3p+(T-1))$$

$$\approx T(1-\alpha-b-c-d)(M-1)$$

For $M \in \{3,4,...\}$ and sufficiently large T the sign of δQ is always positive since $M > 1$ and $1-\alpha-b-c-d > 0$, hence $((0,0,1,0,0),(0,0,1,0,0))$ is a NE ■

Proposition 4.2.4 For sufficiently large T and for $M \in \{3,4,...\}$, the following is a symmetric NE.

$$((0,0,0,1,0),(0,0,0,1,0))$$

Proof As before,

$$Q_1((0,0,0,1,0),(0,0,0,1,0)) =$$

$$\begin{bmatrix}0 & 0 & 0 & 1 & 0\end{bmatrix}\begin{bmatrix}TM & 3T(1-p) & TM & TM & TM\\ 3pT & T & 3p+(T-1) & 3p\log_2 M+(T-\log_2 M) & \frac{T}{2}(1+3p)\\ TM & 3(1-p)+(T-1) & TM & TM & TM\\ TM & 3(1-p)\log_2 M+(T-\log_2 M) & TM & TM & TM\\ TM & \frac{T}{2}(4-3p) & TM & TM & TM\end{bmatrix}\begin{bmatrix}0\\0\\0\\1\\0\end{bmatrix} =$$

$$= \begin{bmatrix}0 & 0 & 0 & 1 & 0\end{bmatrix}\begin{bmatrix}TM\\ 3plog_2M+(T-log_2M)\\ TM\\ TM\\ TM\end{bmatrix} = TM$$

Suppose that P_1 switches to the strategy $\tau = (\alpha, 1-\alpha-b-c-d, b, c, d)$

$Q_1(\tau,(0,0,0,1,0)) =$

$$= \begin{bmatrix}\alpha & 1-\alpha-b-c-d & b & c & d\end{bmatrix} \begin{bmatrix} TM \\ 3plog_2M + (T - log_2M) \\ TM \\ TM \\ TM \end{bmatrix} =$$

$$= \alpha TM + (1-\alpha-b-c-d)(3plog_2M + (T - log_2M)) + bTM + cTM + dTM$$

$\delta Q = Q_1((0,0,0,1,0),(0,0,0,1,0)) - Q_1(\tau,(0,0,0,1,0)) =$

$$= TM - \alpha TM - (1-\alpha-b-c-d)(3plog_2M + (T - log_2M)) - bTM - cTM - dTM$$

$$= TM(1-\alpha-b-c-d) - (1-\alpha-b-c-d)(3plog_2M + (T - log_2M))$$
$$\approx T(1-\alpha-b-c-d)(M-1)$$

For $M \in \{3,4,...\}$ and sufficiently large T the sign of δQ is always positive since $M > 1$ and $1-\alpha-b-c-d > 0$, hence $((0,0,0,1,0),(0,0,0,1,0))$ is a NE point.

Proposition 7.2.5 For sufficiently large T and for $M \in \{3,4,...\}$, the following is a symmetric NE.

$$((0,0,0,0,1),(0,0,0,0,1))$$

Proof As before,

$Q_1((0,0,0,0,1),(0,0,0,0,1)) =$

$$\begin{bmatrix}0 & 0 & 0 & 0 & 1\end{bmatrix} \begin{bmatrix} TM & 3T(1-p) & TM & TM & TM \\ 3pT & T & 3p+(T-1) & 3p\log_2 M + (T - \log_2 M) & \frac{T}{2}(1+3p) \\ TM & 3(1-p)+(T-1) & TM & TM & TM \\ TM & 3(1-p)\log_2 M + (T-\log_2 M) & TM & TM & TM \\ TM & \frac{T}{2}(4-3p) & TM & TM & TM \end{bmatrix} \begin{bmatrix}0\\0\\0\\0\\1\end{bmatrix} =$$

$$= \begin{bmatrix}0 & 0 & 0 & 0 & 1\end{bmatrix} \begin{bmatrix} TM \\ \frac{T}{2}(1+3p) \\ TM \\ TM \\ TM \end{bmatrix} = TM$$

Suppose that P_1 switches to the strategy $\tau = (\alpha, 1-\alpha-b-c-d, b, c, d)$

$Q_1(\tau,(0,0,0,0,1)) =$

$$= \begin{bmatrix}\alpha & 1-\alpha-b-c-d & b & c & d\end{bmatrix} \begin{bmatrix} TM \\ \frac{T}{2}(1+3p) \\ TM \\ TM \\ TM \end{bmatrix} =$$

$$= \alpha TM + (1-\alpha-b-c-d)(\frac{T}{2}(1+3p)) + bTM + cTM + dTM$$

$\delta Q = Q_1((0,0,0,0,1),(0,0,0,0,1)) - Q_1(\tau,(0,0,0,0,1)) =$

$$= TM - \alpha TM - (1-\alpha-b-c-d)(\frac{T}{2}(1+3p)) - bTM - cTM - dTM$$

$$= TM(1-\alpha-b-c-d) - (1-\alpha-b-c-d)(\frac{T}{2}(1+3p))$$

$$= T(1-\alpha-b-c-d)\left(M - \frac{1}{2}(1+3p)\right)$$

For $M \in \{3, 4, ...\}$ and sufficiently large T the sign of δQ is always positive since $M > \frac{1}{2}(1+3p)$ and $1-\alpha-b-c-d > 0$, hence $((0,0,0,0,1),(0,0,0,0,1))$ is a NE ■

Proposition 7.2.6 For $M \in \{4, 8, 16, \dots\dots\}$, and for sufficiently large T, a set of symmetric NE is the following

$$S = \{((\alpha, 0, b, c, 1-\alpha-b-c), (\alpha', 0, b', c', 1-\alpha'-b'-c')) \mid a, b, c \geq 0, \quad \alpha+b+c \leq 1\}$$

This proposition also includes propositions (7.2.1), (7.2.3), (7.2.4) and (7.2.5), since the points they deal with belong to the set S.

Proof From the payoff matrix C, both the column player and the row player obtain the maximum payoff for all pure strategies 1, 3, 4 and 5, when these are played against any combination among them. More specifically, the elements of the payoff matrix satisfy the relation $C_{ij} = TM$ for all $i, j \in \{1, 3, 4, 5\}$.

Therefore, if one player chooses any mixed strategy of the family $x = (a, 0, b, c, 1-a-b-c)$ with $a, b, c \geq 0$ and $a+b+c \leq 1$, the expected payoff that the opponent will also obtain if he plays any combination of strategies 1, 3, 4 or 5 will be exactly TM.

$Q_1(x, x') =$

$$= \begin{bmatrix} a & 0 & b & c & 1-a-b-c \end{bmatrix} \begin{bmatrix} \alpha'TM + b'TM + c \ + (1-a'-b'-c')TM \\ 3pT\alpha' + (3p(T-1))b' + (3plog_2M + T - log_2M)c' + (\frac{T}{2}(1-3p))(1-a'-b'-c') \\ \alpha'TM + b'TM + c \ + (1-a'-b'-c')TM \\ \alpha'TM + b'TM + c \ + (1-a'-b'-c')TM \\ \alpha'TM + b'TM + c \ + (1-a'-b'-c')TM \end{bmatrix} =$$

$$= \alpha'TM + b'TM + c\,TM + (1-a'-b'-c')TM = TM$$

For the profile (x, x') with $x' = (a', 0, b', c', 1-a'-b'-c')$ and $a', b', c' \geq 0$, $a'+b'+c' \leq 1$, to be a Nash equilibrium point, it suffices to show that no player has an incentive to deviate unilaterally.

The only alternative strategy left to examine is strategy 2. The expected payoff of strategy 2 against x is given by the combination:

$Q_1((0,1,0,0,0), x') =$

$$= \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} TM \\ 3pT\alpha' + (3p(T-1))b' + (3plog_2M + T - log_2M)c' + (\frac{T}{2}(1-3p))(1-a'-b'-c') \\ TM \\ TM \\ TM \end{bmatrix} =$$

$$= 3pT\alpha' + (3p(T-1))b' + (3plog_2M + T - log_2M)c' + (\tfrac{T}{2}(1-3p))(1-a'-b'-c')$$

For sufficiently large T we have:

$$Q_1((0,1,0,0,0), x') = T\left(3p\alpha' + 3pb' + c' + (\frac{1}{2}(1-3p))(1-a'-b'-c')\right)$$

$\delta Q = Q_1(x, x') - Q_1((0,1,0,0,0), x') =$

$$= TM - T\left(3p\alpha' + 3pb' + c' + (\frac{1}{2}(1-3p))(1-a'-b'-c')\right)$$

$$= T\left(M - 3p\alpha' - 3pb' - c' - (\frac{1}{2}(1-3p))(1-a'-b'-c')\right)$$

If we prove that for the pure strategies belonging to the set of x' it holds that $\delta Q > 0$, then it is proved that the same holds for all mixed strategies.

Hence:

- For $\alpha' = 1$ we have $M - 3p > 0$, which holds
- For $b' = 1$ we have $M - 3p > 0$, which holds
- For $c' = 1$ we have $M - 1 > 0$, which holds
- For $\alpha' + b' + c' = 0$ we have $M - \frac{1}{2}(1 - 3p) > 0$, which holds

Since the quantity δQ is therefore positive for every x', the row player obviously has no incentive to change strategy. It is proved similarly that the column player does not have an incentive to change strategy either. Consequently the set of points S, as defined in the proposition, constitutes a set of NE ■

Proposition 4.2.7 For $M \in \{3, 4, \dots\}$, for $p \geq \frac{2}{3}$ and for sufficiently large T, the following are symmetric NE points

$$\left(\left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0, 0, 0\right), \left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0, 0, 0\right)\right)$$

$$\left(\left(0, \frac{T(M-1)+1-3p}{T(M-1)-1}, \frac{3p-2}{T(M-1)-1}, 0, 0\right), \left(0, \frac{T(M-1)+1-3p}{T(M-1)-1}, \frac{3p-2}{T(M-1)-1}, 0, 0\right)\right)$$

Proof The proof, since the new strategies have been set to zero, is the same as the one presented in the thesis of Lazaridis-Kechagias (4.3). We look for Nash equilibria of the form

$$\big((a, 1-a, 0), (c, 1-c, 0)\big).$$

In this case the payoff functions are obtained as

$$Q_1(a, c) = TMca + 3Tcp + 2Ta - 2Tac - 3Tap + T - Tc,$$

$$Q_2(a, c) = TMca + 3Tap + 2Tc - 2Tac - 3Tcp + T - Ta.$$

Computing the derivatives

$$\frac{dQ_1}{da} = TMc + 2T - 2Tc - 3Tp,$$

$$\frac{dQ_2}{dlc} = TMa + 2T - 2Ta - 3Tp,$$

and solving the corresponding system of equations, we obtain

$$a = c = \frac{3p-2}{M-2}.$$

These values are admissible only when $p \geq \frac{2}{3}$, so the Nash equilibrium obtained is

$$\left(\left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0\right), \left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0\right)\right)$$

In the same way we look for Nash equilibria of the form $\big((0, a, 1-a), (0, c, 1-c)\big)$, and we have

$$Q_1(a, c) = 2c - 3cp + 3ap + cT - a - ca + aT - aTc + TM - TMc - TMa + cTMa,$$

$$Q_2(a, c) = 2a - 3ap + 3cp + cT - c - ca + aT - aTc + TM - TMc - TMa + cTMa.$$

$$\frac{dQ_1}{da} = 3p - 1 - c + T - Tc - TM + cTM = 0,$$

$$\frac{dQ_2}{dc} = 3p + T - 1 - a - aT - TM + TMa = 0.$$

Solving the above system gives

$$a = c = \frac{TM - T - 3p + 1}{TM - T - 1}.$$

We observe that $TM - T - 1 > 0$. Moreover, $TM - T - 3p + 1 \geq TM - T - 2 = T(M-1) - 2$, which, for sufficiently large values of T, is non-negative. Therefore,

$$a = c \geq 0.$$

It also holds that

$$\frac{TM - T - 3p + 1}{TM - T - 1} \in \left[\frac{TM - T - 2}{TM - T - 1}, \frac{TM - T + 1}{TM - T - 1}\right] \subseteq [0, 1].$$

Hence, since $p \geq \frac{2}{3}$, the point

$$\left(\left(0, \frac{T(M-1) - 3p + 1}{T(M-1) - 1}, \frac{3p - 2}{T(M-1) - 1}\right), \left(0, \frac{T(M-1) - 3p + 1}{T(M-1) - 1}, \frac{3p - 2}{T(M-1) - 1}\right)\right)$$

is a NE ■

Proposition 4.2.8 For $M \in \{3, 4, \dots\}$, for $p \geq \frac{2}{3}$ and for sufficiently large T, the following is a symmetric NE point

$$\left(\left(0, \frac{T(M-1) + (1-3p)log_2 M}{T(M-1) - log_2 M}, 0, \frac{(3p-2)log_2 M}{T(M-1) - log_2 M}, 0\right), \left(0, \frac{T(M-1) + (1-3p)log_2 M}{T(M-1) - log_2 M}, 0, \frac{(3p-2)log_2 M}{T(M-1) - log_2 M}, 0\right)\right)$$

Proof We examine the form $((0, a, 0, 1-a, 0), (0, c, 0, 1-c, 0))$. In a similar way as above we obtain:

$$Q_1(a, c) = acT + a(3plog_2 M + (T - log_2 M))(1 - c) + (1 - a)c(3(1 - p)log_2 M + (T - log_2 M)) + (1 - a)(1 - c)TM$$
$$Q_2(a, c) = acT + a(3plog_2 M + (T - log_2 M))c + (1 - a)(1 - c)(3(1 - p)log_2 M + (T - log_2 M)) + (1 - a)(1 - c)TM$$

and solving

$$\frac{dQ_1}{da} = cT + (3plog_2 M + (T - log_2 M))(1 - c) - c(3(1 - p)log_2 M + (T - log_2 M)) - (1 - c)TM$$
$$\frac{dQ_2}{dc} = aT + (3plog_2 M + (T - log_2 M))(1 - a) - a(3(1 - p)log_2 M + (T - log_2 M)) - (1 - a)TM$$

we get

$$a = c = \frac{T(M-1) + (1-3p)log_2 M}{T(M-1) - log_2 M}$$

Admissible points are those for which the constraints $M \in \{3, 4, \dots\}$ and $p > \frac{2}{3}$ hold. For $p = \frac{2}{3}$ the point is $(0, 1, 0, 0, 0)$, for which we have already proved that it is a symmetric NE ■

Proposition 4.2.9 For $M \in \{3, 4, \dots\}$, for $p \geq \frac{2}{3}$ and for sufficiently large T, the following is a symmetric NE point

$$\left(\left(0, \frac{2M - 3p - 1}{2M - 3}, 0, 0, \frac{3p - 2}{2M - 3}\right), \left(0, \frac{2M - 3p - 1}{2M - 3}, 0, 0, \frac{3p - 2}{2M - 3}\right)\right)$$

Proof We examine the form $((0, a, 0, 0, 1-a), (0, c, 0, 0, 1-c))$. In a similar way as above we obtain:

$$Q_1(a, c) = acT + a(1 - c)\frac{T}{2}(3p + 1) + (1 - a)c\frac{T}{2}(4 - 3p) + (1 - a)(1 - c)TM$$
$$Q_2(a, c) = acT + a(1 - c)\frac{T}{2}(4 - 3p) + (1 - a)c\frac{T}{2}(3p + 1) + (1 - a)(1 - c)TM$$

and solving

$$\frac{dQ_1}{da} = cT + (1-c)\frac{T}{2}(3p+1) - c\frac{T}{2}(4-3p) - (1-c)TM$$
$$\frac{dQ_2}{dc} = aT + (1-c)\frac{T}{2}(3p+1) - c\frac{T}{2}(4-3p) - (1-c)TM$$

we have

$$a = c = \frac{2M-3p-1}{2M-3}$$

Admissible points are those for which the constraints $M \in \{3,4,\dots\}$ and $p > \frac{2}{3}$ hold. For $p = \frac{2}{3}$ the point is $(0,1,0,0,0)$, for which we have already proved that it is a symmetric NE point ■

4.3 Replicator Dynamics

The evolution of the population in this game follows the differential equation below, where x_1 is the fraction of the population following the first strategy, x_2 the second, and so on.

$$\frac{dx_i}{dt} = x_m((Cx)_i - x^T Cx) \tag{7}$$

In more detail:

$$B_1 = C_{11}x_1 + C_{12}x_2 + C_{13}x_3 + C_{14}x_4 + C_{15}x_5$$
$$B_2 = C_{21}x_1 + C_{22}x_2 + C_{23}x_3 + C_{24}x_4 + C_{25}x_5$$
$$B_3 = C_{31}x_1 + C_{32}x_2 + C_{33}x_3 + C_{34}x_4 + C_{35}x_5$$
$$B_4 = C_{41}x_1 + C_{42}x_2 + C_{43}x_3 + C_{44}x_4 + C_{45}x_5$$
$$B_5 = C_{51}x_1 + C_{52}x_2 + C_{53}x_3 + C_{54}x_4 + C_{55}x_5$$

$$\bar{B} = x_1B_1 + x_2B_2 + x_3B_3 + x_4B_4 + x_5B_5$$

$$\frac{dx_1}{dt} = x_1\left(C_{11}x_1 + C_{12}x_2 + C_{13}x_3 + C_{14}x_4 + C_{15}x_5 - \bar{B}\right)$$
$$\frac{dx_2}{dt} = x_2\left(C_{21}x_1 + C_{22}x_2 + C_{23}x_3 + C_{24}x_4 + C_{25}x_5 - \bar{B}\right)$$
$$\frac{dx_3}{dt} = x_3\left(C_{31}x_1 + C_{32}x_2 + C_{33}x_3 + C_{34}x_4 + C_{35}x_5 - \bar{B}\right)$$
$$\frac{dx_4}{dt} = x_4\left(C_{41}x_1 + C_{42}x_2 + C_{43}x_3 + C_{44}x_4 + C_{45}x_5 - \bar{B}\right)$$
$$\frac{dx_5}{dt} = x_5\left(C_{51}x_1 + C_{52}x_2 + C_{53}x_3 + C_{54}x_4 + C_{55}x_5 - \bar{B}\right)$$

To find equilibrium points, the following system must be solved:

$$\forall m : 0 = x_m\left((Cx)_m - x^T Cx\right) \tag{8}$$

The following points are solutions of equation (8), which is why they are examined.

Proposition 4.3.1 For every $M \in \{3,4,\dots\}$ and every T, the following are admissible equilibrium points.

$$\{(x_1, 0, x_3, x_4, 1-x_1-x_3-x_4) \mid x_1, x_3, x_4 \geq 0 \text{ and } x_1+x_3+x_4 \leq 1\}$$

Proof Since they are roots of equation (8) and the necessary constraints for them to represent fractions hold, the points are admissible equilibrium points ■

Proposition 7.3.2 For every $M \in \{3, 4, \dots\}$, $p \geq \frac{2}{3}$ and every T, the following are admissible equilibrium points

$$\left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0, 0, 0\right)$$

$$\left(0, \frac{T(M-1)+1-3p}{T(M-1)-1}, \frac{3p-2}{T(M-1)-1}, 0, 0\right)$$

Proof The proof has been presented in the thesis of Lazaridis-Kechagias. The above are roots of (8) and are feasible for $p \geq \frac{2}{3}$ ■

Proposition 4.3.3 For every $M \in \{3, 4, \dots\}$, $p \geq \frac{2}{3}$ and for sufficiently large T, the following is an admissible equilibrium point

$$\left(0, \frac{T(M-1)+(1-3p)log_2 M}{T(M-1)-log_2 M}, 0, \frac{(3p-2)log_2 M}{T(M-1)-log_2 M}, 0\right)$$

Proof It holds that:

$$\frac{T(M-1)+(1-3p)log_2 M}{T(M-1)-log_2 M} + \frac{(3p-2)log_2 M}{T(M-1)-log_2 M} = 1$$

For sufficiently large T it holds that $T > log_2 M$, so the denominator is positive, as is the numerator of the fraction x_2. Also, for $p \geq \frac{2}{3}$ and since $M > 1$, we have $(3p-2)log_2 M > 0$. Hence this is an admissible equilibrium point ■

Proposition 4.3.4 For every $M \in \{3, 4, \dots\}$, $p \geq \frac{2}{3}$ and every T, the following is an admissible equilibrium point

$$\left(0, \frac{2M-3p-1}{2M-3}, 0, 0, \frac{3p-2}{2M-3}\right)$$

Proof It holds that:

$$\frac{2M-3p-1}{2M-3} + \frac{3p-2}{2M-3} = 1$$

For $M > 3$ it holds that $2M - 3p - 1 > 0$, and obviously $2M - 3 > 0$. Also, for $p \geq \frac{2}{3}$ it holds that $3p - 2 > 0$. Hence the above is an admissible equilibrium point ■

The stability analysis of the above admissible equilibrium points follows. The eigenvalues of the Jacobian matrix of the differentials of equation (7) at each point were used.

Proposition 4.3.5 Regarding the set of equilibrium points

$$S = \{(x_1, 0, x_3, x_4, 1 - x_1 - x_3 - x_4) \mid x_1, x_3, x_4 \geq 0 \text{ and } x_1 + x_3 + x_4 \leq 1\}$$

For $M \in \{3, 4, \dots\}$ and for sufficiently large T, it is a stable non-isolated set of equilibria.

Proof The Jacobian has eigenvalues

$$\lambda_1 = 0$$
$$\lambda_2 = 0$$
$$\lambda_3 = 0$$
$$\lambda_4 = -MT$$
$$\lambda_5 = \frac{T}{2} - x_3 - MT + \frac{3Tp}{2} - \frac{Tx_1}{2} + \frac{Tx_3}{2} + \frac{Tx_4}{2} - \log_2(M)x_4 + 3px_3 + \frac{3Tpx_1}{2} - \frac{3Tpx_3}{2} - \frac{3Tpx_4}{2} + 3\log_2(M)px_4$$

It is proven that the non-zero eigenvalues are strictly negative. Since all eigenvalues are non-positive but three of them are zero, to determine stability we must examine the system in more detail.

Suppose we perturb the system from an arbitrary equilibrium point $(x_1^*, 0, x_3^*, x_4^*, x_5^*)$ of the set S, by introducing a small population $\epsilon_2 > 0$ in strategy 2. For the sum of the populations to remain equal to 1, the remaining strategies undergo small decreases δ_i (where $\sum \delta_i = \epsilon_2$). The new state is $(x_1^* - \delta_1, \epsilon_2, x_3^* - \delta_3, x_4^* - \delta_4, x_5^* - \delta_5)$.

Based on the Jacobian matrix, the linearized dynamics for the critical perturbation away from the surface S reduce to:

$$\frac{d\epsilon_2}{dt} = \lambda_5 \epsilon_2 \tag{9}$$

where $\lambda_5 < 0$. The dynamics along the surface S (directions x_1, x_3, x_4, x_5) correspond to the zero eigenvalues, that is, their perturbations are not self-reinforcing.
Since the sum of the changes must be zero ($\sum \frac{dx_i}{dt} = 0$), the dynamics of the remaining neutral strategies will collectively be:

$$\frac{dx_1}{dt} + \frac{dx_3}{dt} + \frac{dx_4}{dt} + \frac{dx_5}{dt} = -\lambda_5 \epsilon_2 \tag{10}$$

If the system undergoes this perturbation, we observe from equation (9) that the perturbation ϵ_2 will decrease exponentially to 0 (since $\lambda_5 < 0$). At the same time, according to equation (10), the populations of strategies 1, 3, 4 and 5 will readjust, essentially absorbing the decreasing population ϵ_2, until the moment when ϵ_2 becomes practically zero.

Consequently, the system will not return exactly to the initial point $(x_1^*, 0, x_3^*, x_4^*, x_5^*)$. Instead, it will end up at a new equilibrium point of the form $(x_1', 0, x_3', x_4', x_5')$.

In short, while the individual points of S are stable, the whole set $S = \{(x_1, 0, x_3, x_4, x_5) : \sum x_i = 1\}$ is a stable non-isolated set of equilibria ■

Proposition 4.3.6 Regarding the equilibrium point

$$\left(0, \frac{T(M-1) + (1-3p)log_2 M}{T(M-1) - log_2 M}, 0, \frac{(3p-2)log_2 M}{T(M-1) - log_2 M}, 0\right)$$

For $M \in \{3, 4, \dots\}$, $p \geq \frac{2}{3}$ and for sufficiently large T, the equilibrium point is unstable.

Proof We find the eigenvalues:

$$\lambda_1 = \frac{(3p-2)(\log_2(M)-1)(T(M-1) + \log_2(M)(1-3p))}{T(M-1) - \log_2(M)}$$

$$\lambda_2 = \frac{\log_2(M)(3p-2)(T(M-1) + \log_2(M)(1-3p))}{T(M-1) - \log_2(M)}$$

$$\lambda_3 = -\frac{T\log_2(M) - MT^2 + 9p\log_2^2(M) + T^2 - 2\log_2^2(M) - 9p^2\log_2^2(M)}{T + \log_2(M) - MT}$$

$$\lambda_4 = -\frac{(3p-2)(T - 2\log_2(M))(T - \log_2(M) - MT + 3p\log_2(M))}{T + \log_2(M) - MT}$$

$$\lambda_5 = -\frac{(3p-2)(T - 2\log_2(M))(T - \log_2(M) - MT + 3p\log_2(M))}{2(T + \log_2(M) - MT)}$$

Regarding the eigenvalue λ_1, for $p \geq \frac{2}{3}$ and for sufficiently large T, we have proved that the numerator is a positive number, as is the denominator. Hence, since $\lambda_1 > 0$, the above equilibrium point is unstable ■

Proposition 4.3.7 Regarding the equilibrium point

$$\left(0, \frac{2M-3p-1}{2M-3}, 0, 0, \frac{3p-2}{2M-3}\right)$$

For $M \in \{3, 4, \dots\}$, $p \geq \frac{2}{3}$ and for sufficiently large T, the equilibrium point is unstable.

Proof We find the eigenvalues:

$$\lambda_1 = -\frac{(3p-2)(T-2\log_2(M))(3p-2M+1)}{4M-6}$$

$$\lambda_2 = -\frac{T(9p^2-9p+4M-4)}{4M-6}$$

$$\lambda_3 = -\frac{(3p-2)(T-2)(3p-2M+1)}{4M-6}$$

$$\lambda_4 = -\frac{T(3p-2)(3p-2M+1)}{4M-6}$$

$$\lambda_5 = -\frac{T(3p-2)(3p-2M+1)}{4M-6}$$

Regarding the eigenvalue λ_3, for $p \geq \frac{2}{3}$ and for sufficiently large T, we have proved that $3p-2M+1 < 0$, so the numerator is a negative number and the denominator is positive. Hence $\lambda_3 > 0$, so the above equilibrium point is unstable ■

Proposition 4.3.8 Regarding the equilibrium point

$$\left(0, \frac{T(M-1)+1-3p}{T(M-1)-1}, \frac{3p-2}{T(M-1)-1}, 0, 0\right)$$

For $M \in \{3, 4, \dots\}$, $p \geq \frac{2}{3}$ and for sufficiently large T, the equilibrium point is unstable.

Proof We find the eigenvalues:

$$\lambda_1 = \frac{(3p-2)(T+3p-MT-1)}{T-MT+1}$$

$$\lambda_2 = -\frac{T+9p-MT^2+T^2-9p^2-2}{T-MT+1}$$

$$\lambda_3 = -\frac{(3p-2)(T-1)(T+3p-MT-1)}{T-MT+1}$$

$$\lambda_4 = -\frac{(3p-2)(\log_2(M)-1)(T+3p-MT-1)}{T-MT+1}$$

$$\lambda_5 = -\frac{(3p-2)(T-2)(T+3p-MT-1)}{2(T-MT+1)}$$

Regarding the eigenvalue λ_1, for $p \geq \frac{2}{3}$ and for sufficiently large T, we have proved that the numerator is a negative number and the denominator positive. Hence $\lambda_1 > 0$, so the above equilibrium point is unstable ■

Proposition 4.3.9 Regarding the equilibrium point

$$\left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0, 0, 0\right)$$

For $M \in \{5, 6, ...\}$, $p > \frac{2}{3}$ and for sufficiently large T, the equilibrium point is unstable.

Proof We find the eigenvalues:

$$\lambda_1 = \frac{(3p-2)(M-3p)(T-1)}{M-2}$$

$$\lambda_2 = \frac{T(3p-2)(M-3p)}{2(M-2)}$$

$$\lambda_3 = \frac{(3p-2)(M-3p)(T-\log_2(M))}{M-2}$$

$$\lambda_4 = -\frac{T(9p^2-9p+M)}{M-2}$$

$$\lambda_5 = \frac{T(3p-2)(M-3p)}{M-2}$$

We examine the numerators of the eigenvalues, since the denominator is obviously positive:

- For $p > \frac{2}{3}$ we have $3p-2 > 0$, so at least one eigenvalue is a strictly positive number, hence the equilibrium point is unstable.
- For $p < \frac{2}{3}$ we have $3p-2 < 0$, so $\frac{3p-2}{M-2} < 0$, hence it is not an admissible point ■

Proposition 4.3.10 Regarding the equilibrium point

$$(0, 1, 0, 0, 0)$$

For $M \in \{3, 4, \dots\}$, for $p > \frac{2}{3}$ ($p < \frac{2}{3}$) and for sufficiently large T, the equilibrium point is asymptotically stable (unstable).

Proof We find the eigenvalues:

$$\lambda_1 = -(3p-2), \qquad \lambda_2 = -T(3p-2), \qquad \lambda_3 = -\log_2(M)(3p-2), \qquad \lambda_4 = -T, \qquad \lambda_5 = -\frac{T(3p-2)}{2}$$

When the quantity $3p-2 > 0$, all eigenvalues are negative. Consequently the above equilibrium point is asymptotically stable when $p > \frac{2}{3}$ and unstable for $p < \frac{2}{3}$ ■

4.4 Evolutionarily Stable Strategies

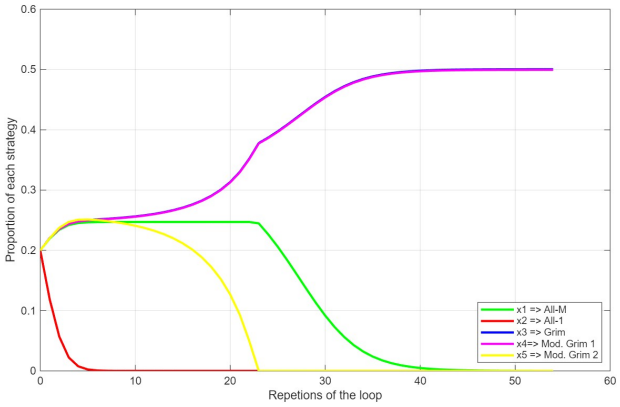
The following propositions characterize the symmetric Nash equilibrium points of the five-strategy game with respect to their evolutionary stability.

Proposition 4.4.1 The set of Nash equilibrium points

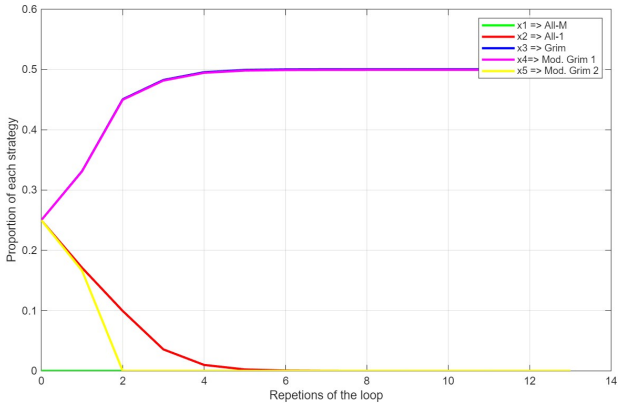
$$S = \{(x_1, 0, x_3, x_4, 1-x_1-x_3-x_4) \mid x_1, x_3, x_4 \geq 0 \text{ and } x_1+x_3+x_4 \leq 1\}$$

is a set of ESS.

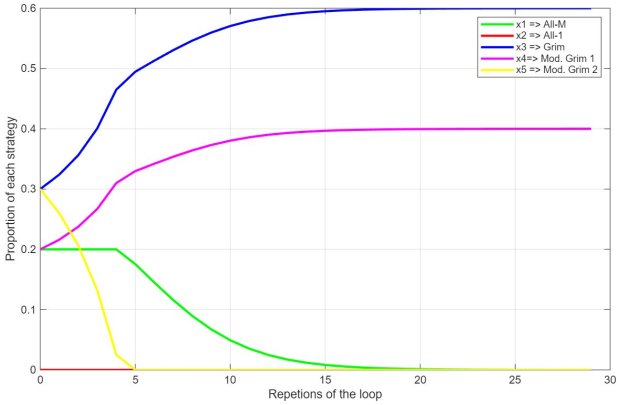
Proof We take any strategy of the set $\hat{\mathbf{x}} = (x_1, 0, x_3, x_4, x_5) \in S$ (where $x_5 = 1-x_1-x_3-x_4$) and $\mathbf{x} = (y_1, y_2, y_3, y_4, y_5)$. From the construction of the updated payoff matrix C, we know that for every $i, j \in \{1, 3, 4, 5\}$ it holds that $C_{ij} = MT$. Therefore, the vector of expected payoffs $C\hat{\mathbf{x}}$ is particularly simple.:



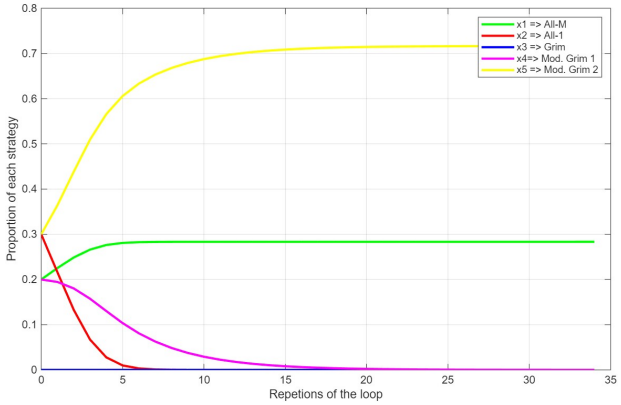
(a) $M = 8, T = 10, p = 0.75, x = [0.2, 0.2, 0.2, 0.2, 0.2]$



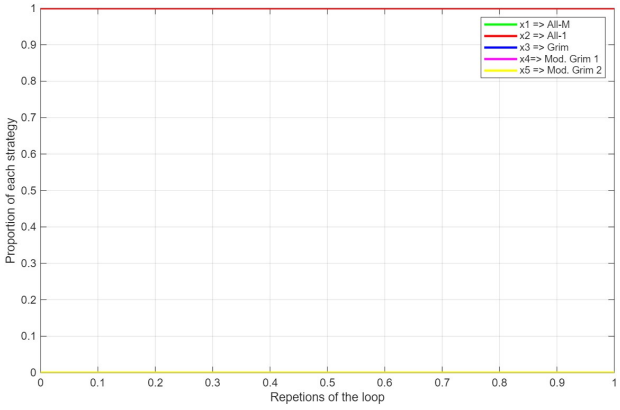
(b) $M = 8, T = 10, p = 0.75, x = [0, 0.25, 0.25, 0.25, 0.25]$



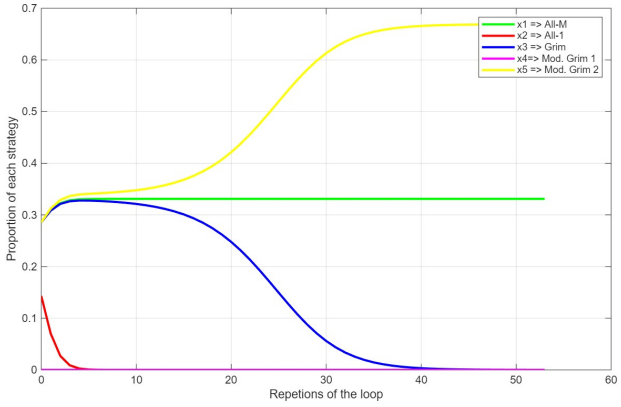
(c) $M = 8, T = 10, p = 0.75, x = [0.2, 0, 0.3, 0.2, 0.3]$



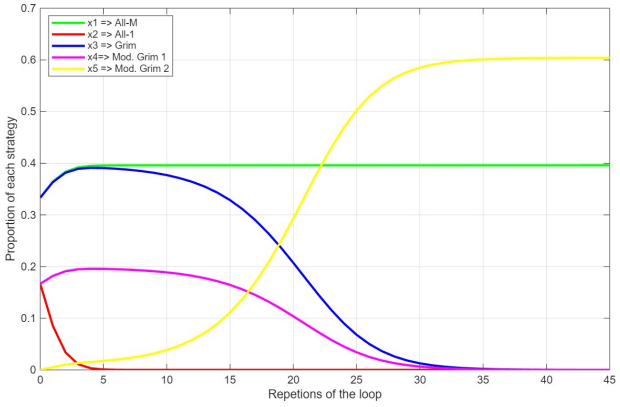
(d) $M = 8, T = 10, p = 0.75, x = [0.2, 0.3, 0, 0.2, 0.3]$



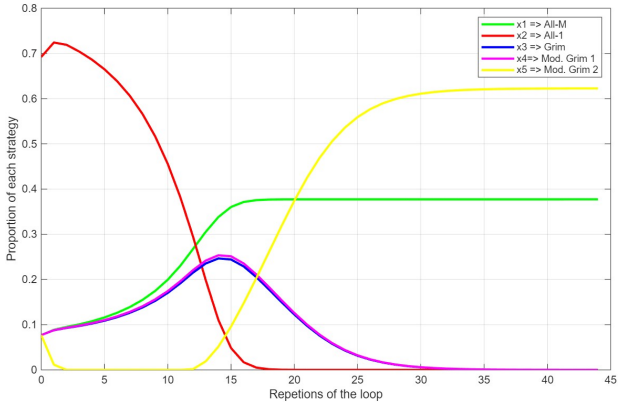
(e) $M = 8, T = 10, p = 0.75, x = [0, 1, 0, 0, 0]$



(f) $M = 8, T = 10, p = 0.75, x = [\frac{2}{7}, \frac{1}{7}, \frac{2}{7}, 0, \frac{2}{7}]$



(g) $M = 8, T = 10, p = 0.75, x = [\frac{1}{3}, \frac{1}{6}, \frac{1}{3}, \frac{1}{6}, 0]$



(h) $M = 8, T = 10, p = 0.75, x = [\frac{1}{13}, \frac{9}{13}, \frac{1}{13}, \frac{1}{13}, \frac{1}{13}]$

Figure 8: Replicator Dynamics experiments with 5 strategies for various initial conditions

$$C\hat{\mathbf{x}} = \begin{bmatrix} TM \\ (C\hat{\mathbf{x}})_2 \\ TM \\ TM \\ TM \end{bmatrix}$$

where:

$$(C\hat{\mathbf{x}})_2 = 3pTx_1 + (3p + T - 1)x_3 + (3p\log_2 M + T - \log_2 M)x_4 + \frac{T}{2}(1 + 3p)x_5$$

Hence we have:

$$\hat{\mathbf{x}}^T C\hat{\mathbf{x}} = x_1(TM) + 0 \cdot (C\hat{\mathbf{x}})_2 + x_3(TM) + x_4(TM) + x_5(TM) = TM(x_1 + x_3 + x_4 + x_5) = TM$$
$$\mathbf{x}^T C\hat{\mathbf{x}} = y_1(TM) + y_2(C\hat{\mathbf{x}})_2 + y_3(TM) + y_4(TM) + y_5(TM)$$

Since the fractions of $\mathbf{x}$ sum to 1, we have $y_1 + y_3 + y_4 + y_5 = 1 - y_2$. Hence:

$$\mathbf{x}^T C\hat{\mathbf{x}} = TM(1 - y_2) + y_2(C\hat{\mathbf{x}})_2 = TM - y_2(TM - (C\hat{\mathbf{x}})_2)$$

Consequently, the payoff difference is:

$$\hat{\mathbf{x}}^T C\hat{\mathbf{x}} - \mathbf{x}^T C\hat{\mathbf{x}} = TM - [TM - y_2(TM - (C\hat{\mathbf{x}})_2)] = y_2(TM - (C\hat{\mathbf{x}})_2)$$

For sufficiently large T, the sign of this difference is determined by the first-order terms in T, that is, by the parenthesis:

$$TM - (C\hat{\mathbf{x}})_2 \approx TM - T\left(3px_1 + x_3 + x_4 + \frac{1 + 3p}{2}x_5\right) = T\left(M - (3px_1 + x_3 + x_4 + \frac{1 + 3p}{2}x_5)\right)$$

Since x_1, x_3, x_4, x_5 form a probability distribution (fractions), the maximum possible value of the term $(3px_1 + x_3 + x_4 + \frac{1+3p}{2}x_5)$ equals $3p$ (since $p \in (1/2, 1)$ and the coefficient $3p$ is the largest). Hence:

$$T\left(M - (3px_1 + x_3 + x_4 + \frac{1 + 3p}{2}x_5)\right) \geq T(M - 3p)$$

For every $M \geq 3$, $M > 3p$ holds, so we have $T(M - 3p) > 0$. This confirms that the payoff of strategy 2 is strictly smaller than MT. Since the quantity $(TM - (C\hat{\mathbf{x}})_2)$ is therefore strictly positive, we have:

$$\hat{\mathbf{x}}^T C\hat{\mathbf{x}} - \mathbf{x}^T C\hat{\mathbf{x}} = y_2(TM - (C\hat{\mathbf{x}})_2) \geq 0 \implies y_2 \geq 0$$

This inequality obviously holds, since y_2 expresses a population fraction. Equality will hold only when $y_2 = 0$.Conversely, for every strategy $\mathbf{x}$ with $y_2 > 0$, the following will strictly hold:

$$\hat{\mathbf{x}}^T C\hat{\mathbf{x}} > \mathbf{x}^T C\hat{\mathbf{x}}$$

Consequently the set S is a set of ESS ■

Proposition 4.4.2 For $p > \frac{2}{3}$ the Nash equilibrium point

$$\hat{\mathbf{x}} = (0, 1, 0, 0, 0)$$

is an ESS.

Proof We take any strategy $\mathbf{x} = (x_1, x_2, x_3, x_4, x_5)$, with $x_1 + x_2 + x_3 + x_4 + x_5 = 1$. We have:

$$C\hat{\mathbf{x}} = \begin{bmatrix} 3T(1 - p) \\ T \\ 3(1 - p) + (T - 1) \\ 3(1 - p)\log_2 M + (T - \log_2 M) \\ \frac{T}{2}(4 - 3p) \end{bmatrix} \implies \hat{\mathbf{x}}^T C\hat{\mathbf{x}} = T$$

and

$$\mathbf{x}^T C\hat{\mathbf{x}} = 3Tx_1(1-p) + Tx_2 + [T+2-3p]x_3$$
$$+ [T + (2-3p)\log_2 M]\, x_4 + \frac{Tx_5}{2}(4-3p)$$

Therefore, we have the difference

$$\hat{\mathbf{x}}^T C\hat{\mathbf{x}} - \mathbf{x}^T C\hat{\mathbf{x}} = T - 3Tx_1(1-p) - Tx_2 - [T+2-3p]x_3$$
$$-[T + (2-3p)\log_2 M]x_4 - \frac{Tx_5}{2}(4-3p)$$

Separating the $O(T)$ and $O(1)$ terms, we have

$$\hat{\mathbf{x}}^T C\hat{\mathbf{x}} - \mathbf{x}^T C\hat{\mathbf{x}} = T\left[1 - 3(1-p)x_1 - x_2 - x_3 - x_4 - \frac{1}{2}(4-3p)x_5\right]$$
$$+ [(2-3p)x_3 - (2-3p)\log_2 Mx_4]$$

Analyzing the $O(T)$ terms, we have

$$1 - 3(1-p)x_1 - x_2 - x_3 - x_4 - \frac{1}{2}(4-3p)x_5 =$$

$$x_1 + x_5 - 3(1-p)x_1 - \frac{1}{2}(4-3p)x_5 =$$

$$x_1(3p-2) + x_5\left(-1 + \frac{3p}{2}\right) =$$

$$(3p-2)\left(x_1 + \frac{x_5}{2}\right)$$

For $p > \frac{2}{3}$, the term $(3p-2)$ is strictly positive, hence

$$\hat{\mathbf{x}}^T C\hat{\mathbf{x}} - \mathbf{x}^T C\hat{\mathbf{x}} \approx (3p-2)\left(x_1 + \frac{x_5}{2}\right) \geq 0$$

with equality holding only when $x_1 = x_5 = 0$. Conversely, for every $\mathbf{x}$ with $x_1, x_5 > 0$, it strictly holds that

$$\hat{\mathbf{x}}^T C\hat{\mathbf{x}} > \mathbf{x}^T C\hat{\mathbf{x}}$$

Therefore, the NE $(0,1,0,0,0)$ is an ESS for $p > \frac{2}{3}$ ■

Proposition 7.4.3 The Nash equilibrium points

$$\left(0, \frac{T(M-1) + (1-3p)log_2 M}{T(M-1) - log_2 M}, 0, \frac{(3p-2)log_2 M}{T(M-1) - log_2 M}, 0\right),$$

$$\left(0, \frac{2M-3p-1}{2M-3}, 0, 0, \frac{3p-2}{2M-3}\right),$$

$$\left(0, \frac{T(M-1)+1-3p}{T(M-1)-1}, \frac{3p-2}{T(M-1)-1}, 0, 0\right),$$

$$\text{For } p > \frac{2}{3} \text{ then : } \left(\frac{3p-2}{M-2}, \frac{M-3p}{M-2}, 0, 0, 0\right)$$

$$\text{For } p < \frac{2}{3} \text{ then : } (0,1,0,0,0)$$

are not ESS.

Proof These points are unstable equilibrium points, therefore they cannot be evolutionarily stable strategies ■

4.5 Markov Analysis

For the Markov analysis with five strategies, the definitions of the population vector z, the state/strategy vector s, and the frequency vector x remain the same as in the thesis of Lazaridis-Kechagias, with the difference that s and x now consist of 5 elements instead of 3. The protocol followed in the evolutionary game also remains exactly as defined before.

For the strategies that were chosen, the new **Payoff** matrix is obtained as

$$Q(\mathbf{s}) = \begin{bmatrix} (s_1-1)C_{11} + s_2C_{12} + s_3C_{13} + s_4C_{14} + s_5C_{15} \\ s_1C_{21} + (s_2-1)C_{22} + s_3C_{23} + s_4C_{24} + s_5C_{25} \\ s_1C_{31} + s_2C_{32} + (s_3-1)C_{33} + s_4C_{34} + s_5C_{35} \\ s_1C_{41} + s_2C_{42} + s_3C_{43} + (s_4-1)C_{44} + s_5C_{45} \\ s_1C_{51} + s_2C_{52} + s_3C_{53} + s_4C_{54} + (s_5-1)C_{55} \end{bmatrix}$$

$$Q(\mathbf{s}) = \begin{bmatrix} (s_1-1)TM + s_2\,3T(1-p) + s_3TM + s_4TM + s_5TM \\ s_1\,3pT + (s_2-1)T + s_3\big(3p + (T-1)\big) + s_4\big(3p\log_2 M + (T-\log_2 M)\big) + s_5\,\frac{T}{2}(1+3p) \\ s_1TM + s_2\big(3(1-p) + (T-1)\big) + (s_3-1)TM + s_4TM + s_5TM \\ s_1TM + s_2\big(3(1-p)\log_2 M + (T-\log_2 M)\big) + s_3TM + (s_4-1)TM + s_5TM \\ s_1TM + s_2\,\frac{T}{2}(4-3p) + s_3TM + s_4TM + (s_5-1)TM \end{bmatrix}$$

Having moved to a 5-dimensional space, it is not possible to depict the Markov dynamics in three-dimensional space. However, it is feasible to compute the transition matrix P, which will be sparse and square of dimensions $\Omega \times \Omega$, where

$$\Omega = \binom{N+4}{4}$$

The probability $\rho_{i\to j}$ that a player using strategy i switches to strategy j (destination) is based on the payoff surplus ΔQ_{ij}, on the basis of which he examines whether strategy j performs better than his own.

$$\Delta Q_{ij} = \max(0, Q_j - Q_i)$$

The quantity ΔQ_{ij} determines the probability of switching from a strategy i to a strategy j, which is given by the relation

$$\rho_{i\to j} = \frac{\frac{s_j}{N} \cdot \max(0, Q_j - Q_i)}{\sum_{k=1}^{K} \frac{s_k}{N} \cdot \max(0, Q_k - Q_i)}$$

Obviously, if there is no better strategy, that is, all surpluses are ≤ 0, the player stays in place ($\rho_{i\to i} = 1$).

The matrix P describes the probability that the overall state of the population changes from $\vec{s}$ to $\vec{s}'$ in one time step. According to the protocol of the evolutionary game, only one player can revise his strategy at each time step. Thus, the transition probability $P_{\vec{s}\to\vec{s}'}$ is computed on the basis of the following probabilities:

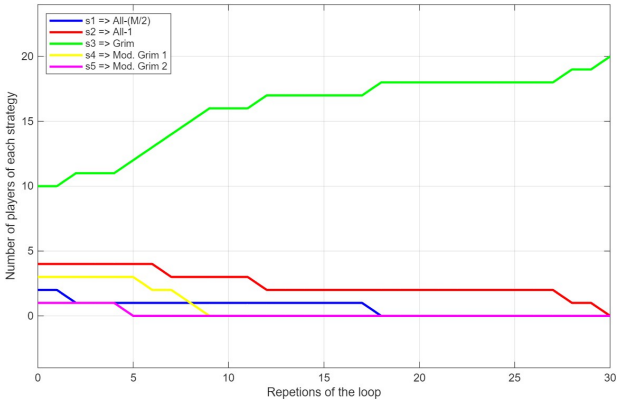
- Player selection probability: The probability that a player playing strategy i is selected for revision, since the selection is made with equal probability, is

$$\frac{s_i}{N}$$

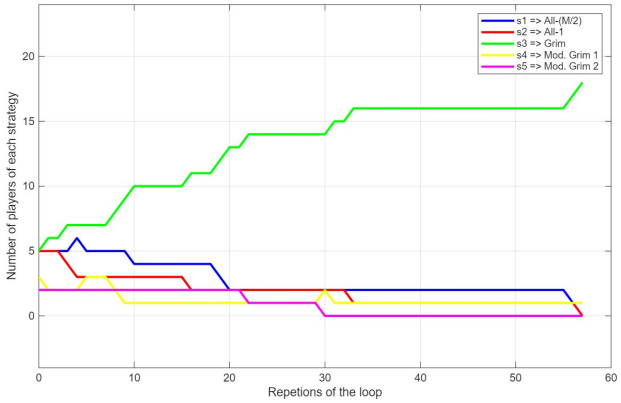
- Switching probability: The probability that this player switches to strategy j is $\rho_{i\to j}$

Since the transition from state s to state s' arises as the intersection of the above events, which are independent of each other, we have

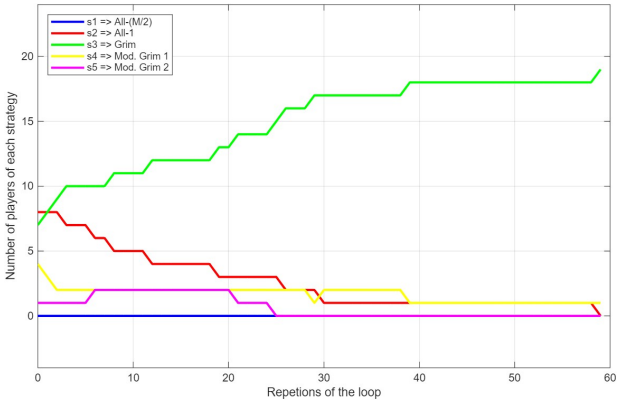
$$P_{i\to j} = \frac{s_i}{N} \cdot \rho_{i\to j}$$



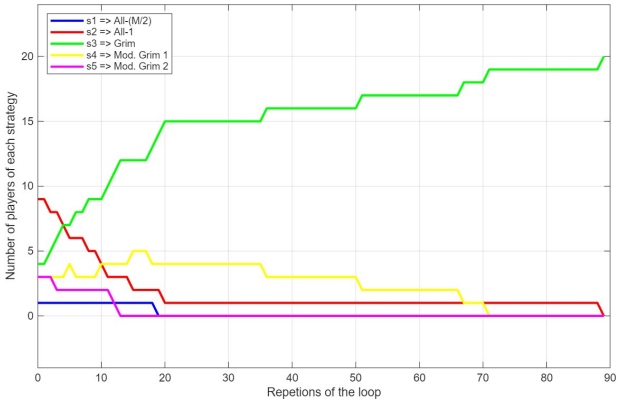
(a) $M = 4, T = 10, p = 0.75, N = 20, s = [2, 4, 10, 3, 1]$



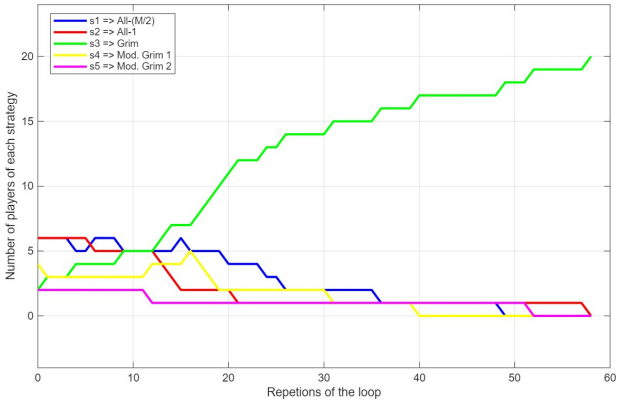
(b) $M = 4, T = 10, p = 0.75, N = 20, s = [5, 5, 5, 3, 2]$



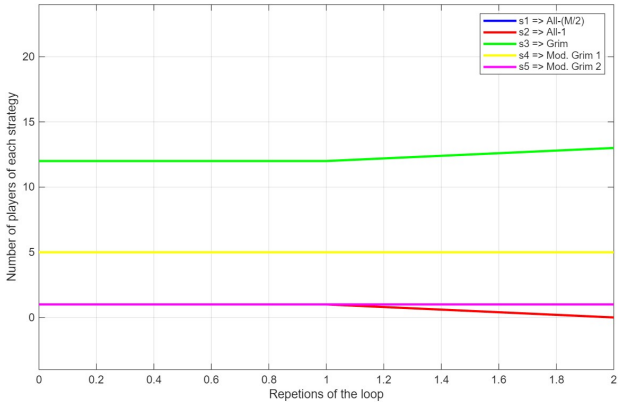
(c) $M = 4, T = 10, p = 0.75, N = 20, s = [0, 8, 7, 4, 1]$



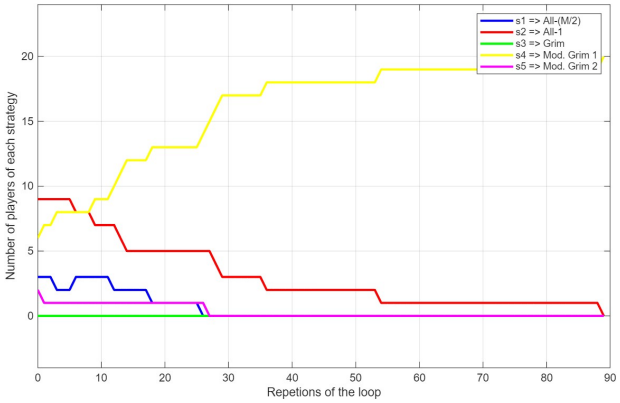
(d) $M = 4, T = 10, p = 0.75, N = 20, s = [1, 9, 4, 3, 3]$



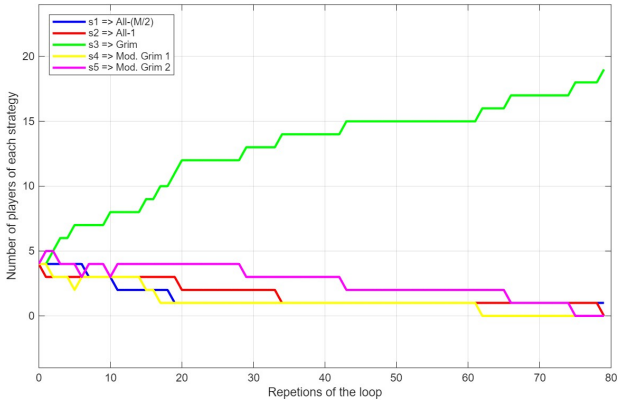
(e) $M = 4, T = 10, p = 0.75, N = 20, s = [6, 6, 2, 4, 2]$



(f) $M = 4, T = 10, p = 0.75, N = 20, s = [1, 1, 12, 5, 1]$



(g) $M = 4, T = 10, p = 0.75, N = 20, s = [3, 9, 0, 6, 2]$



(h) $M = 4, T = 10, p = 0.75, N = 20, s = [4, 4, 4, 4, 4]$

Figure 9: Markov experiments for various initial conditions

5 Two-Population Replicator Dynamics

The replicator dynamics, as presented in Section 3, assumes the existence of a single population of players. Consequently (as can be seen quite easily), the single-population replicator dynamics can only lead to a symmetric Nash equilibrium. However, as we also saw in Section 3, the ISC also has asymmetric Nash equilibrium points. These equilibria can be recovered if we assume that two populations of players play against each other. That is, we always choose one player from Population 1 and the other player from Population 2. In this case, let $x(t)$ (respectively $y(t)$) be the strategy frequency vector of Population 1 (respectively Population 2) at time t; we can assume that the frequencies evolve according to the Two-Population Replicator Dynamics:

$$\frac{dx_m}{dt} = x_m \left((Cy)_m - x^T Cy\right) \tag{11}$$

$$\frac{dy_m}{dt} = y_m \left((x^T C)_m - x^T Cy\right) \tag{12}$$

For the study of this game, the three strategies used in the thesis of Lazaridis-Kechagias are used. Therefore, the payoff matrix C is

$$C = \begin{bmatrix} TM & 3(1-p)T & TM \\ 3pT & T & 3p + (T-1) \\ TM & 3(1-p) + (T-1) & TM \end{bmatrix}$$

All the asymmetric solutions of the system

$$\frac{dx_m}{dt} = 0$$
$$\frac{dy_m}{dt} = 0$$

are examined, and whether they are admissible, that is, whether they express probability fractions.

Proposition 5.1 For every $M \in \{3, 4, \dots\}$ and every T, the following is **not** an admissible equilibrium point.

$$\mathbf{x} = \left[0, \frac{-T + 3p + MT - 2}{MT - T - 1}, \frac{1 - 3p}{MT - T - 1}\right]$$

$$\mathbf{y} = \left[0, \frac{-T - 3p + MT + 1}{MT - T - 1}, \frac{3p - 2}{MT - T - 1}\right]$$

Proof It holds that $MT - T - 1 > 0$, so for the above quantities to express probability fractions, the numerators must be positive numbers. Regarding the population $\mathbf{x}$, x_3 will always be negative since $3p - 1 < 0,\ \forall p \in (\frac{1}{2}, 1]$. Hence this particular solution is not admissible ■

Proposition 5.2 For every $M \in \{3, 4, \dots\}$ and every T, the following is **not** an admissible equilibrium point.

$$\mathbf{x} = \left[\frac{1 - 3p}{M - 2}, \frac{M + 3p - 3}{M - 2}, 0\right]$$

$$\mathbf{y} = \left[\frac{3p - 2}{M - 2}, \frac{M - 3p}{M - 2}, 0\right]$$

Proof It holds that $M - 2 > 0$, so with the same logic as in Proposition **5.1** we want the numerators to be positive quantities. As for x_1 , $1 - 3p < 0$, $\forall p \in (\frac{1}{2}, 1]$, hence the solution is not admissible ■

Proposition 5.3 For every $M \in \{3, 4, \dots\}$ and sufficiently large T, the following is **not** an admissible equilibrium point.

$$\mathbf{x} = \left[\frac{1-3p}{MT-3p-3T+3Tp+1}, \frac{T(M+3p-3)}{MT-3p-3T+3Tp+1}, 0\right]$$

$$\mathbf{y} = \left[0, -\frac{T+3p-MT-1}{MT-3p-3T+3Tp+1}, \frac{T(3p-2)}{MT-3p-3T+3Tp+1}\right]$$

Proof It holds that $TM-3p-3T+3Tp+1>0$, since for sufficiently large T the denominator is equivalent to $T(M-3+3p)>0\ , \forall M \in \{3,4,...\}$. With the same logic as above, $1-3p<0\ , \forall p \in (\frac{1}{2}, 1]$, hence the solution is not admissible ■

Proposition 5.4 For every $M \in \{3, 4, \dots\}$ and every T, the following are admissible equilibrium points

$$\mathbf{x} = [a, 0, 1-a]\,, \qquad \mathbf{y} = [b, 0, 1-b]$$

$$\mathbf{x} = [0, 1, 0]\,, \qquad \mathbf{y} = [1, 0, 0]$$

$$\mathbf{x} = [0, 1, 0]\,, \qquad \mathbf{y} = [0, 0, 1]$$

Proof The above proposition holds since these are solutions of the system. Obviously, admissible solutions also include the fractions where the population vector $\mathbf{x}$ is replaced by the strategies of the population $\mathbf{y}$ and vice versa ■

For the admissible asymmetric equilibrium points, stability will be examined according to the Jacobian matrix of each point and its eigenvalues.

Proposition 5.5 Regarding the set of equilibrium points $(\mathbf{x}, \mathbf{y})$

$$\mathbf{x} = [x_1, 0, 1-x_1]$$

$$\mathbf{y} = [y_1, 0, 1-y_1]$$

For $M \in \{3, 4, \dots\}\ , x_1, y_1 \in [0, 1]$ and for sufficiently large T, the set of points is a stable non-isolated set of equilibria.

Proof We find the eigenvalues:

$$\begin{aligned}
&\lambda_1 = 0\\
&\lambda_2 = 0\\
&\lambda_3 = -MT\\
&\lambda_4 = -MT\\
&\lambda_5 = T - 2x_1 - 3p - MT + 2Tx_1 + 3x_1p - 3Tx_1p + 2\\
&\lambda_6 = T + y_1 + 3p - MT - Ty_1 - 3y_1p + 3Ty_1p - 1
\end{aligned}$$

Obviously $\lambda_3, \lambda_4 < 0$. Regarding the eigenvalue λ_5, for sufficiently large T we have:

$$\lambda_5 \approx T - MT + 2Tx_1 - 3Tx_1p = -T((M-1) + x_1(3p-2)) < 0$$

which holds since

$$M - 1 + x_1(3p-2) > 0 \iff M > 1 + x_1(2-3p), \quad \text{which holds since} \quad \max(x_1(2-3p)) < 0.5$$

Regarding the eigenvalue λ_6, for sufficiently large T we have:

$$\lambda_6 \approx T - MT - Ty_1 + 3Ty_1p = -T(M - 1 + y_1(1-3p)) < 0$$

which holds since

$$M - 1 + y_1(1-3p) > 0 \iff M > 1 + y_1(3p-1) \quad \text{which holds since} \quad max(y_1(3p-1)) = 2$$

Hence all eigenvalues are non-positive, but since at least one of them is equal to zero, to determine stability we must examine the system in more detail.

Suppose we perturb the system from an arbitrary equilibrium point $(\mathbf{x}^*, \mathbf{y}^*)$ of the surface, where $\mathbf{x}^* = (x_1^*, 0, x_3^*)$ and $\mathbf{y}^* = (y_1^*, 0, y_3^*)$, by introducing into the two populations small fractions $\epsilon_2^x > 0$ and $\epsilon_2^y > 0$ in strategy 2. For the sum of the fractions in each population to remain equal to 1, strategies 1 and 3 undergo small decreases δ_i^x and δ_i^y. The new state is $\mathbf{x} = (x_1^* - \delta_1^x, \epsilon_2^x, x_3^* - \delta_3^x)$ and $\mathbf{y} = (y_1^* - \delta_1^y, \epsilon_2^y, y_3^* - \delta_3^y)$.
Based on the Jacobian matrix, the linearized dynamics for the critical perturbations away from the equilibrium surface reduce to the equations:

$$\frac{d\epsilon_2^x}{dt} = \lambda_5 \epsilon_2^x \tag{13}$$

$$\frac{d\epsilon_2^y}{dt} = \lambda_6 \epsilon_2^y \tag{14}$$

where $\lambda_5, \lambda_6 < 0$. The dynamics along the surface (directions x_1, x_3 and y_1, y_3) correspond to the zero eigenvalues (λ_1, λ_2), that is, their perturbations are not self-reinforcing.
Since the sum of the changes in each population must always be zero ($\sum \frac{dx_i}{dt} = 0$ and $\sum \frac{dy_i}{dt} = 0$), the dynamics of the remaining neutral strategies will collectively be:

$$\frac{dx_1}{dt} + \frac{dx_3}{dt} = -\lambda_5 \epsilon_2^x \tag{15}$$

$$\frac{dy_1}{dt} + \frac{dy_3}{dt} = -\lambda_6 \epsilon_2^y \tag{16}$$

If the system undergoes this perturbation, we observe from equations (13) and (14) that: the perturbations ϵ_2^x and ϵ_2^y will decrease until they reach 0 (since $\lambda_5, \lambda_6 < 0$). At the same time, according to equations (15) and (16), the populations of strategies 1 and 3 (for both players) will readjust, absorbing the decreasing ϵ_2^x and ϵ_2^y, until the moment when these become practically zero.

Consequently, the system will not return exactly to the initial point $(\mathbf{x}^*, \mathbf{y}^*)$. Instead, it will end up at a new equilibrium point of the same form, say $(\mathbf{x}', \mathbf{y}') = ((x_1', 0, x_3'), (y_1', 0, y_3'))$.

In short, while the individual equilibrium points are not asymptotically stable, the set of these points $S = \{(\mathbf{x}, \mathbf{y}) \mid x_2 = 0, y_2 = 0\}$ is a stable non-isolated set of equilibria ■

Proposition 5.6 Regarding the equilibrium points $(\mathbf{x}, \mathbf{y})$

$$\mathbf{x} = [0, 1, 0]$$

$$\mathbf{y} = [1, 0, 0]$$

$$\mathbf{x} = [1, 0, 0]$$

$$\mathbf{y} = [0, 1, 0]$$

For $M \in \{3, 4, \dots\}$ and for every large T, the equilibrium points are unstable.

Proof We find the eigenvalues for the first point:

$$\begin{aligned}
\lambda_1 &= T(M-3p)\\
\lambda_2 &= T(M-3p)\\
\lambda_3 &= -T(3p-1)\\
\lambda_4 &= -3Tp\\
\lambda_5 &= -3Tp\\
\lambda_6 &= -(3p-1)(T-1)
\end{aligned}$$

Since at least one eigenvalue is positive ($\lambda_1 = \lambda_2 = T(M-3p) > 0$), the equilibrium point is unstable. Instability is proved similarly for the second equilibrium point ■

Proposition 5.7 Regarding the equilibrium points $(\mathbf{x}, \mathbf{y})$

$$\mathbf{x} = [0,1,0]\,, \qquad \mathbf{y} = [0,0,1]$$

$$\mathbf{x} = [0,0,1]\,, \qquad \mathbf{y} = [0,1,0]$$

For $M \in \{3,4,\dots\}$ and for every T, the equilibrium points are unstable.

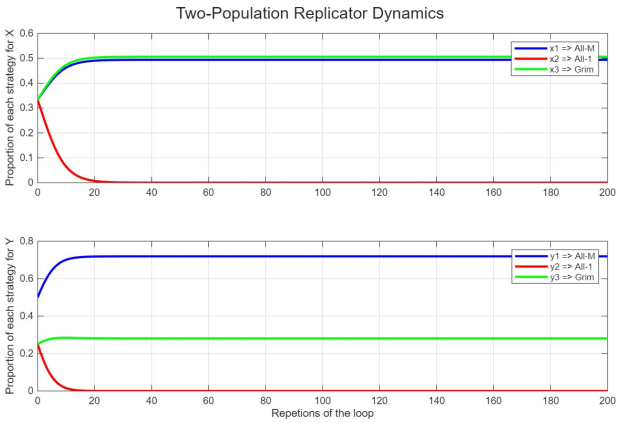
Proof We find the eigenvalues for the first point:

$$\begin{aligned}
\lambda_1 &= 1-3p\\
\lambda_2 &= 1-3p-T\\
\lambda_3 &= 1-3p-T\\
\lambda_4 &= (3p-1)(T-1)\\
\lambda_5 &= MT-3p-T+1\\
\lambda_6 &= MT-3p-T+1
\end{aligned}$$

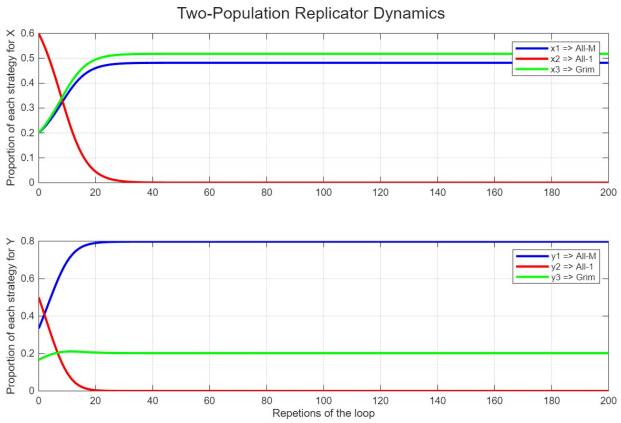
Since at least one eigenvalue is positive ($\lambda_4 = (3p-1)(T-1) > 0$), the equilibrium point is unstable. Instability is proved similarly for the second equilibrium point ■

6 Conclusion

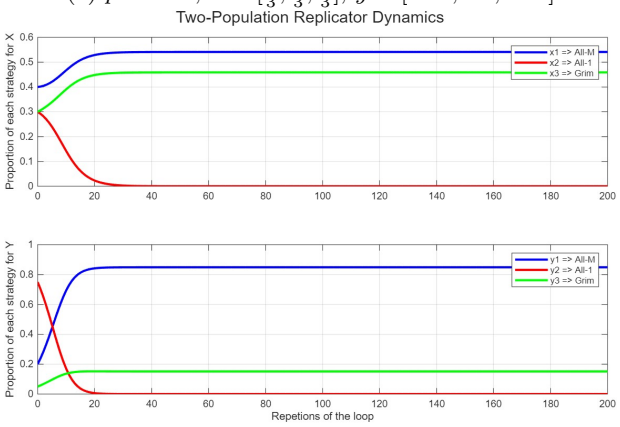
In this work, we extended the work of Lazaridis-Kechagias, taking fundamental mechanisms, such as Markov chains and replicator dynamics, and re-implementing them when creating the five-strategy system. We found that the results of the simulations, as well as the central theoretical concepts, such as the Nash equilibrium, have significant social implications, particularly in the context of promoting cooperation and punishing defection. Overall, the knowledge acquired offers analytical tools with broad application in a multitude of other scientific fields, going beyond the limits of game theory.



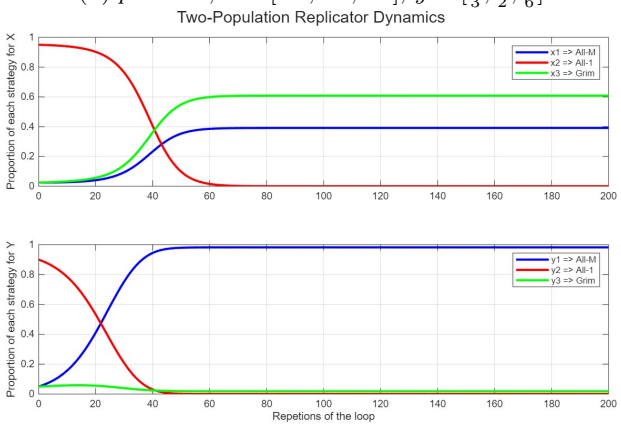
(a) $p = 0.75$, $x = [\frac{1}{3}, \frac{1}{3}, \frac{1}{3}]$, $y = [0.25, 0.5, 0.25]$



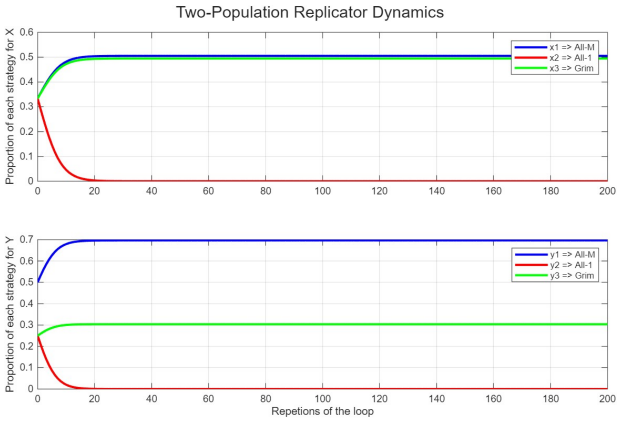
(b) $p = 0.75$, $x = [0.2, 0.6, 0.2]$, $y = [\frac{1}{3}, \frac{1}{2}, \frac{1}{6}]$



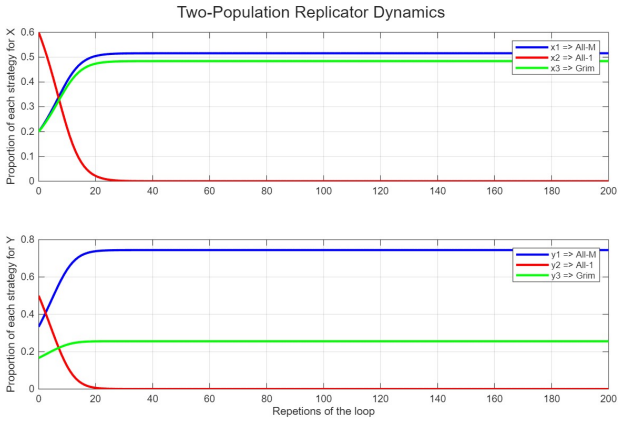
(c) $p = 0.75$, $x = [0.4, 0.3, 0.3]$, $y = [0.2, 0.75, 0.05]$



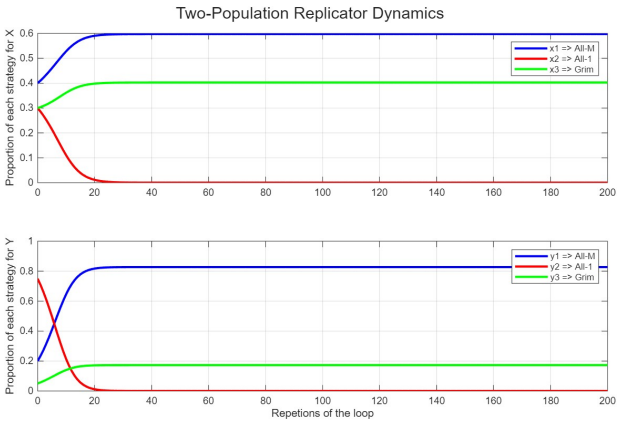
(d) $p = 0.75$, $x = [0.025, 0.95, 0.025]$, $y = [0.05, 0.9, 0.05]$



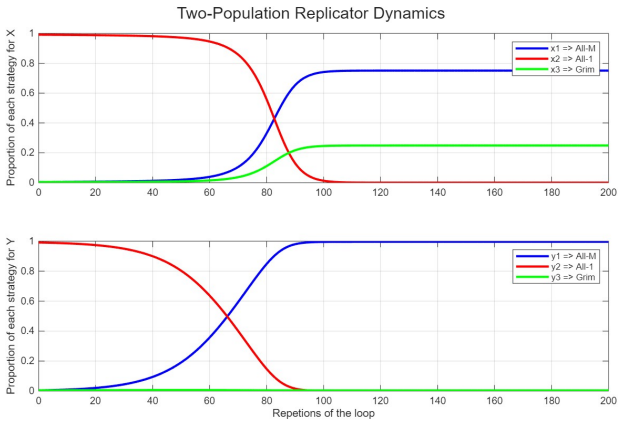
(e) $p = 0.6$, $x = [\frac{1}{3}, \frac{1}{3}, \frac{1}{3}]$, $y = [0.25, 0.5, 0.25]$



(f) $p = 0.6$, $x = [0.2, 0.6, 0.2]$, $y = [\frac{1}{3}, \frac{1}{2}, \frac{1}{6}]$



(g) $p = 0.6$, $x = [0.4, 0.3, 0.3]$, $y = [0.2, 0.75, 0.05]$



(h) $p = 0.6$, $x = [\frac{1}{232}, \frac{115}{116}, \frac{1}{232}]$, $y = [\frac{1}{232}, \frac{115}{116}, \frac{1}{232}]$

Figure 10: Two-Population Replicator experiments for various initial conditions, with $M = 4$, $T = 10$ and $T_f = 1$

7 Appendix

7.1 Implementation and Analysis of 3 Strategies

- **RepDyn.m**

  The function simulates the evolution of the ISC for the matrix C as presented in Section 3, over time. Depending on the parameters p, M, T and the initial conditions x_0, the simulation is run either for a specific time T_f or until the system reaches equilibrium, and finally the results of the simulation are plotted on a set of axes.

- **PhasePlot.m**

  The function plots a phase diagram that visualizes the change over time of the fractions of the three different strategies according to the system of differential equations (1), given the values of the parameters p, M, and T.

- **Q_matrix.m**

  The function computes a vector Q that represents the difference between the expected payoff of each strategy and the mean payoff of the population, given the number of individuals (s_1, s_2, s_3) choosing each strategy. These values determine the trends of change of the population in the discrete evolution model, based on the payoff matrix C of the game of Section 3

- **state2node.m**

  The function converts a particular population state $s = [s_1, s_2, s_3]$ (where the sum of individuals is N) into a unique integer index corresponding to the node n. It is used for the linear numbering and mapping of all possible combinations of the population triangle-simplex into a two-dimensional structure.

- **node2state.m**

  The function performs the inverse operation of the previous one, that is, it converts a node number back into the corresponding population state $s = [s_1, s_2, s_3]$ for total population N.

- **StateTransitionGraph.m**

  The function performs two operations:

  – It returns the matrix P, which has dimension $\frac{(N+1)(N+2)}{2} \times \frac{(N+1)(N+2)}{2}$ and

  $$P_{ij} : \text{Probability of transition from state } i \text{ to state } j.$$

  Each state (s_1, s_2, s_3) is mapped to a number $1, 2, ..., \frac{(N+1)(N+2)}{2}$ inside the function, and in this way both the probabilities of strategy change and the probabilities of retaining them (self-loops) are captured.

  – It visualizes the Markov dynamics through a directed graph, where

  * Each node corresponds to a state (s_1, s_2, s_3)
  * Two nodes are connected by a directed edge when the probability of transition from the state corresponding to the first to the state corresponding to the second is non-zero (omitting self-loops)
  * The color of each node corresponds to an RGB triplet, which visualizes the long-run probability of transition to each of the three vertices of the graph, with:
    · **R**: the long-run probability of transition to the state $(N, 0, 0)$
    · **G**: the long-run probability of transition to the state $(0, N, 0)$
    · **B**: the long-run probability of transition to the state $(0, 0, N)$

- **color_nodes.m**

  The function computes a color representation (in RGB form) for each node of the Markov lattice, based on the long-run convergence probabilities (limiting distribution) of the Markov chain. Specifically, it raises the transition matrix P to a very large power (P^{1000}) to simulate the steady state of the system over time and isolates the probabilities of ending up in the three "pure" vertices of the triangle: All-(M/2) (R), All-1 (G) and Grim (B). Finally, it normalizes these values to the interval $[0, 1]$ so that they can be used as valid RGB colors for visually highlighting the basins of attraction of each strategy on the graph.

- **MarkDyn.m**

  The function is an analogous implementation of **RepDyn.m**, adapted however to the context of Markov processes, that is, to discrete states corresponding to a specific number of individuals. Its operation is based on the transition probability matrix P and, through an iterative loop of duration T_m, it simulates the evolution of

the game under examination. In this context, the dynamics of the system are now probabilistic, since at every moment each individual has a certain probability of switching from one strategy to another, according to the probabilities defined by the matrix P.

- **xRep001.m, xRep002.m, ... , xRep005.m**

  These scripts aim to study the replicator dynamics for different initial distributions of the populations. Each of them initializes the game parameters (M, T, p), generates either a random or a specific valid initial population distribution (x_0) and calls the function RepDyn to simulate it, while at the same time plotting the phase diagram of the strategies through PhasePlot.

- **xMark001.m, ... , xMark004.m**

  These scripts aim to study the Markovian dynamics for different initial distributions in a discrete and finite population of size N. Each of them constructs the transition probability matrix of the system through the function StateTransitionGraph and then simulates the evolution of the population states over time with the function MarkDyn, starting from a specified initial distribution of individuals (s_0).

7.2 Implementation and Analysis of 5 Strategies

- **RepDynFive.m**

  The function first constructs the updated matrix C, as presented in Section 5. It then computes the mean payoff of the entire population (B), numerically solves the system of 5 differential equations with the Euler method and produces a summary time plot of the evolution of all strategies.

- **GenerateStateSpace.m**

  The function creates and maps the set of all possible population states for a game of 5 strategies with a fixed total population of N players. More specifically, it computes the number of possible combinations (nchoosek(N + 4, 4)) and uses 4 nested loops to find all valid distributions of players among the 5 strategies ($s_1+s_2+s_3+s_4+s_5 = N$). Finally, it returns a matrix states with all population triples/quintuples and a multidimensional lookup array (state_lookup), which allows direct retrieval of the unique identifier of a state based on the frequencies of its strategies.

- **Q_matrix_5.m**

  The function extends the operation of Q_matrix to 5 strategies, computing the differences between the expected payoffs of each strategy and the mean payoff of the population. It takes as input the number of players $(s_1, \ldots, s_5)$ following each strategy and uses the payoff matrix C of Section 5 to return the corresponding vector Q of change trends.

- **StateTransitionsFive.m**

  The function constructs the transition probability matrix P (of dimensions $S \times S$, where S is the number of states) for the Markov dynamics of 5 strategies. For each population state, it computes a revision protocol based on the tendency to imitate the most profitable strategies, determining the probability that the system moves to a neighboring state due to the strategy change of one and only one player.

- **MarkDynFive.m**

  The function simulates the Markov evolution of the 5-strategy system over time, starting from an initial population distribution s_0. At each step, it uses a random number generator and the cumulative probabilities of the matrix P to choose the next state, while at the end it plots the evolution over time of the number of players for each of the 5 strategies.

- **xRepFive001.m, ... , xRepFive008.m**

  What xRep001.m, ... , xRep005.m are for Section 3

- **xMarkFive001.m, ... , xMarkFive008.m**

  What xMark001.m, ... , xMark004.m are for Section 3

7.3 Two-Population Replicator Dynamics

- **RepDyn2Pop.m**

  The function simulates the Two-Population Replicator Dynamics for the asymmetric game with the matrix C presented in the thesis of Lazaridis-Kechagias, where the two players come from different evolutionary sets (X and Y). It computes the expected payoff of the interaction of the two populations (xCy), numerically solves the system of 6 differential equations (3 for each population) and produces a double plot that separately visualizes the evolution over time of the strategies (All-M, All-1, Grim) for the population of Players 1 (X) and of Players 2 (Y).

- **xTwoPopRep001.m, ... , xTwoPopRep008.m**

  Each of these scripts sets the fixed parameters of the game (M, T, p, T_f), initializes and normalizes the different initial frequency distributions for the first population and the second population, and finally calls the function RepDyn2Pop to run the simulation and draw the plots.